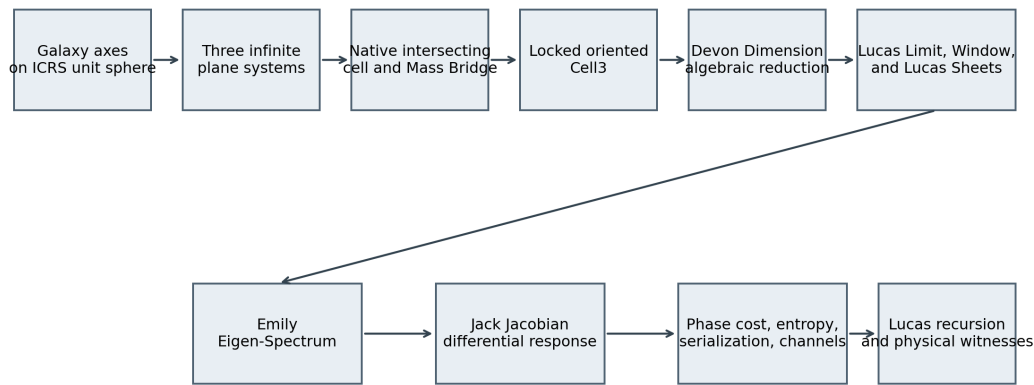


K-Field: A Student Introduction

Galaxy-Axis Plane Systems, Cell3, Devon, Lucas, Emily, Jack, Phase Structure, and Recursive Physical Organization

Typed K-Field dependency architecture



K-Field -> Oriented Cell3 -> Devon -> Lucas Limit -> Lucas Window -> Lucas Sheets -> Emily; Jack acts across the hierarchy.

The complete typed dependency architecture, including the galaxy-axis unit-sphere root, the three infinite plane systems, the Mass-Bridge shape lock, the Devon algebraic layer, the Lucas minimum-complete domain, the Emily spectral layer, and the Jack differential operator.

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Abstract

The K-Field is the physically established, oriented geometric substrate that organizes direction, phase, polarity, metastable resolution, reciprocal structure, and exact recursive growth across physical systems. Its primitive local unit is Cell3, an oblique three-dimensional metric-address cell with a fixed similarity shape, a locked orientation in the International Celestial Reference System, a 27-state centered ternary registry, seven alternating-polarity sheets, and thirteen antipodal projective channels. The Mass Bridge fixes the Cell3 projective face-area point through exact relationships to the proton-electron mass ratio and the fine-structure channel, leaving uniform positive scale as the only intrinsic continuous freedom. Direct and reciprocal metrics describe physical displacements and plane systems; a quadratic phase metric assigns orientation-dependent transition cost; path multiplicity supplies the entropy of metastable resolution; and a lossless scalar serialization supplies a directed address that is demonstrably nonlocal relative to physical adjacency. Cell3 fills volume through the Lucas hierarchy. A $3 \times 3 \times 3$ array forms a Lucas Group, while a $9 \times 9 \times 9$ array forms a Lucas Supergroup containing 729 Cell3 volumes, 6,859 refined states, 55 Devon Super Sheets, and an exact four-channel recursive morphology. Physical witness layers connect the fixed geometry to mass, atomic, electromagnetic, gravitational, and Planck-family relationships. This treatment develops the complete framework from first principles, distinguishes deterministic geometry from probabilistic outcome, and states the present experimental and dynamical implications in a form suitable for foundational university study.

The derivation begins with Earth-observed galaxy-axis orientations expressed in ICRS and projected onto an antipodally completed unit sphere. Independent geometric methods recover three infinite plane systems with fixed ICRS orientations and unit-sphere-normalized spacings. Their intersections define a native oblique cell. The Mass Bridge then selects the exact projective face-area point and leaves uniform scale as the only intrinsic continuous freedom. The resulting oriented Cell3 is reduced into the Devon Dimension, expanded at the Lucas Limit into the 27-node Lucas Window, partitioned into actual Lucas Sheets, diagonalized by the Emily Eigen-Spectrum, and differentiated by the Jack Jacobian. These named structures are typed stages of one physical hierarchy rather than interchangeable labels.

The K-Field provides the total structure; Cell3 provides the geometry; Devon reduces it; the Lucas Limit defines completeness; the Lucas Window realizes it; the Lucas Sheets organize it; Emily diagonalizes it; and Jack describes how it changes.

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1. The K-Field as a Physical Framework

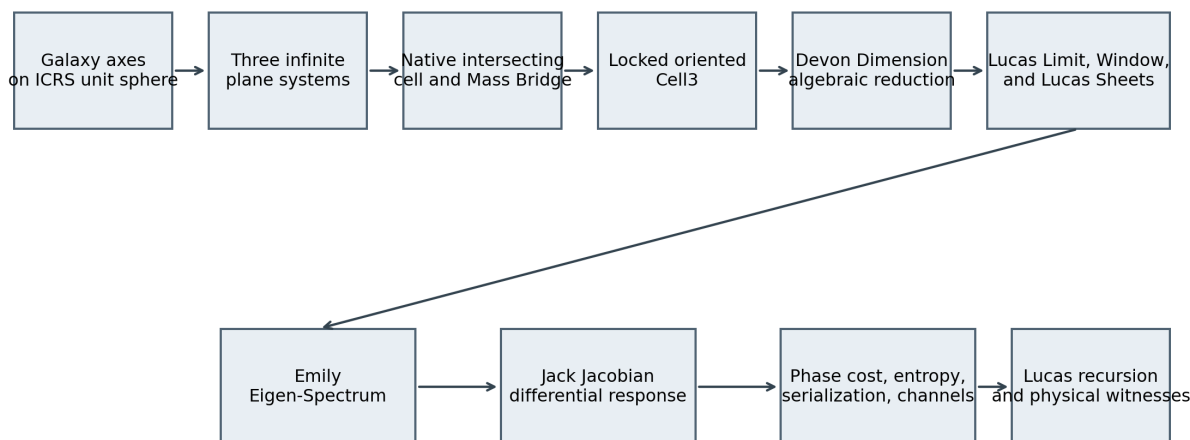
The K-Field is a universal oriented substrate rather than a single conventional force. It defines the geometrical and informational conditions under which physical systems possess directions, addresses, phase costs, polarities, histories, and recursive relations. A force law describes how a system accelerates under specified conditions. The K-Field begins earlier: it specifies the structured state space in which those conditions and responses are formed.

The canonical K-Field object contains a fixed ICRS origin and orientation, an oblique basis B , a direct metric, a reciprocal metric, an integer address lattice, a laminar grade, a checkerboard polarity, an affine three-dimensional incidence structure, and a projective Mass Bridge. An observer packet may add time, velocity, orientation, coupling strength, and apparatus response. The distinction between the substrate and the observer packet is essential. The substrate is fixed; the observer moves and changes orientation relative to it.

The physical content is distributed across several mutually compatible representations. Direct space describes displacement. Reciprocal space describes plane normals and wavevectors. The 27-state registry describes local incidence. The seven-sheet complex describes laminar grade and polarity. The 13-channel projective system describes antipodal directions and metric reconstruction. Scalar serialization describes directed address order. Path entropy describes multiplicity of histories. Lucas recursion describes volume-filling closure.

No single representation replaces the others. The same state may be geometrically near another state, far away in scalar rank, incident with several projective lines, and located on a Devon Super Sheet of fixed polarity. Each relation answers a different physical question. Confusion results when distance, address, incidence, probability, and phase cost are treated as interchangeable.

Typed K-Field dependency architecture



K-Field -> Oriented Cell3 -> Devon -> Lucas Limit -> Lucas Window -> Lucas Sheets -> Emily; Jack acts across the hierarchy.

Figure 1. The complete typed dependency architecture, including the galaxy-axis unit-sphere root, the three infinite plane systems, the Mass-Bridge shape lock, the Devon algebraic layer, the Lucas minimum-complete domain, the Emily spectral layer, and the Jack differential operator.

The controlling named progression

The internal hierarchy is K-Field -> Oriented Cell3 -> Devon Dimension -> Lucas Limit -> Lucas Window -> Lucas Sheet Decomposition -> Emily Eigen-Spectrum. The Jack Jacobian acts across the hierarchy as its differential response operator. The order is a dependency order. Geometry must exist before it can be reduced algebraically; the algebraic rules must exist before a finite window can evaluate them; the graph must exist before its spectrum can be diagonalized; and a derivative map cannot replace the unperturbed object that it differentiates.

Established status and open dynamics

Within the K-Field research program, the field effect is experimentally established and Cell3 is the mathematically locked primitive geometry. Exact combinatorial, projective, spectral, differential, and recursive results follow from the locked structure. The remaining theoretical frontier is the compact dynamical action governing exchange of energy, momentum, phase, and entropy with matter.

2. Galaxy-Axis Unit Sphere and the Infinite Three-Plane System

The observational geometric root is an Earth-observed catalog of galaxy-axis orientations expressed in the International Celestial Reference System. A galaxy axis is an unoriented line: reversing its direction does not create a different physical axis. Each catalog axis is therefore represented by an antipodal pair on the unit sphere.

Analysis of the antipodally completed unit-sphere distribution recovers three associated K-Field plane systems. Each system consists of an infinite family of parallel planes. A family is specified by a unit normal and an interplanar spacing. The three normals are not mutually orthogonal, so the corresponding cell is oblique rather than cubic.

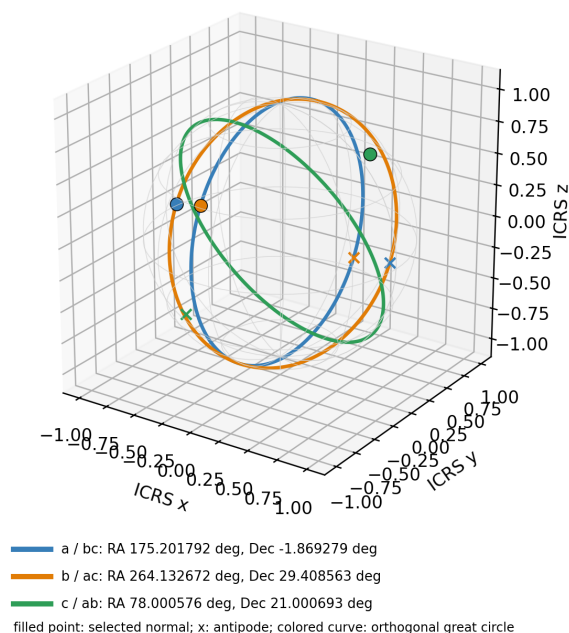
The locked plane-family normals are the normals to the bc, ac, and ab faces of Cell3. Their ICRS coordinates are $(-0.995965190309, 0.083602162507, -0.032619290564)$, $(-0.089052594102, -0.866576769291, 0.491033948326)$, and $(0.194092200808, 0.913177168821, 0.358379234791)$. Their right ascensions and declinations are 175.201792 deg, -1.869279 deg; 264.132672 deg, 29.408563 deg; and 78.000576 deg, 21.000693 deg, respectively.

The corresponding opposite-face spacings are 0.329546529695 nm, 0.451597379721 nm, and 0.605097741519 nm in the adopted scaled realization. The full plane systems extend without boundary. The central Cell3 is one primitive region between adjacent members of all three families.

The dependency chain distinguishes the native galaxy-derived intersecting cell from the exact locked Cell3. The plane systems supply the observed orientation architecture and an initial geometric reconstruction. A minute perturbation bridge preserves that source while the Mass Bridge enforces the exact projective face-area closure. The final Cell3 therefore retains the galaxy-axis plane-system origin and orientation while occupying the unique Mass-Bridge similarity class.

Galaxy-axis unit-sphere extraction and the three K-Field plane systems

Recovered plane-system normals on the galaxy-axis unit sphere



Finite view of three infinite parallel-plane stacks and central Cell3

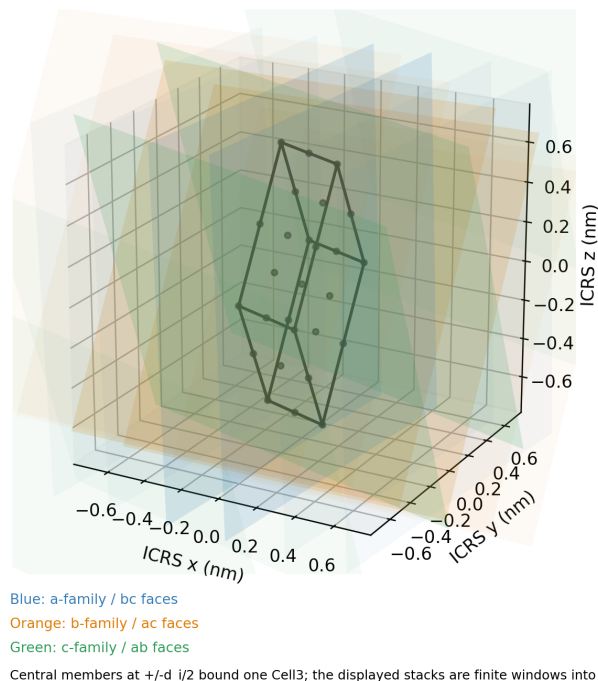


Figure 2. The observational geometric root. The left panel places the three recovered plane-system normals and their orthogonal great circles on the antipodally interpreted ICRS unit sphere. The right panel shows finite windows into the corresponding infinite parallel-plane systems; adjacent central members bound the Cell3 volume.

Plane equations

Let g_i be the reciprocal covectors. The infinite centered boundary-plane families are

$$Pi_i(m): g_i \cdot (x - O_{ICRS}) = m + 1/2, \quad m \in \mathbb{Z}.$$

The central Cell3 is the simultaneous region

$$-1/2 \leq g_a \cdot (x - O_{ICRS}) \leq 1/2$$

$$-1/2 \leq g_b \cdot (x - O_{ICRS}) \leq 1/2$$

$$-1/2 \leq g_c \cdot (x - O_{ICRS}) \leq 1/2.$$

3. Cell3: The Primitive Ontological Unit

Cell3 is the primitive metric-address unit of the K-Field. It is ontological because its intrinsic geometry is treated as part of the physical constitution of the substrate rather than as a coordinate decoration or a material crystal assembled by ordinary matter. Matter samples and responds to Cell3 structure; it does not create that structure.

The locked basis matrix B has columns a, b, and c. These columns generate an oblique parallelepiped. The basis is left-handed in its recorded ICRS representation, with signed determinant -0.11791489191174465 nm^3 and positive volume 0.11791489191174465 nm^3. Handedness records orientation; volume records the absolute scale of the displayed realization.

The complete ontological object is affine, not merely metric. A metric specifies lengths and angles but is unchanged by translation and rotation of the entire cell. The K-Field additionally specifies an ICRS embedding, a lattice phase, and a discrete translation grammar. The resulting structure can distinguish apparatus orientation and, when phase is measured, translational position relative to the lattice origin.

Locked numerical geometry

Quantity	Value
a	0.334179679059 nm
b	0.586410890411 nm
c	0.792319765043 nm
theta_ab	83.933483012313 deg
theta_ac	80.448130018338 deg
theta_bc	50.363231813756 deg
volume	0.117914891912 nm^3
longest body diagonal	1.343653621676 nm
handedness	left-handed in recorded ICRS representation

Locked Cell3: 27 numbered nodes and six primitive boundary planes

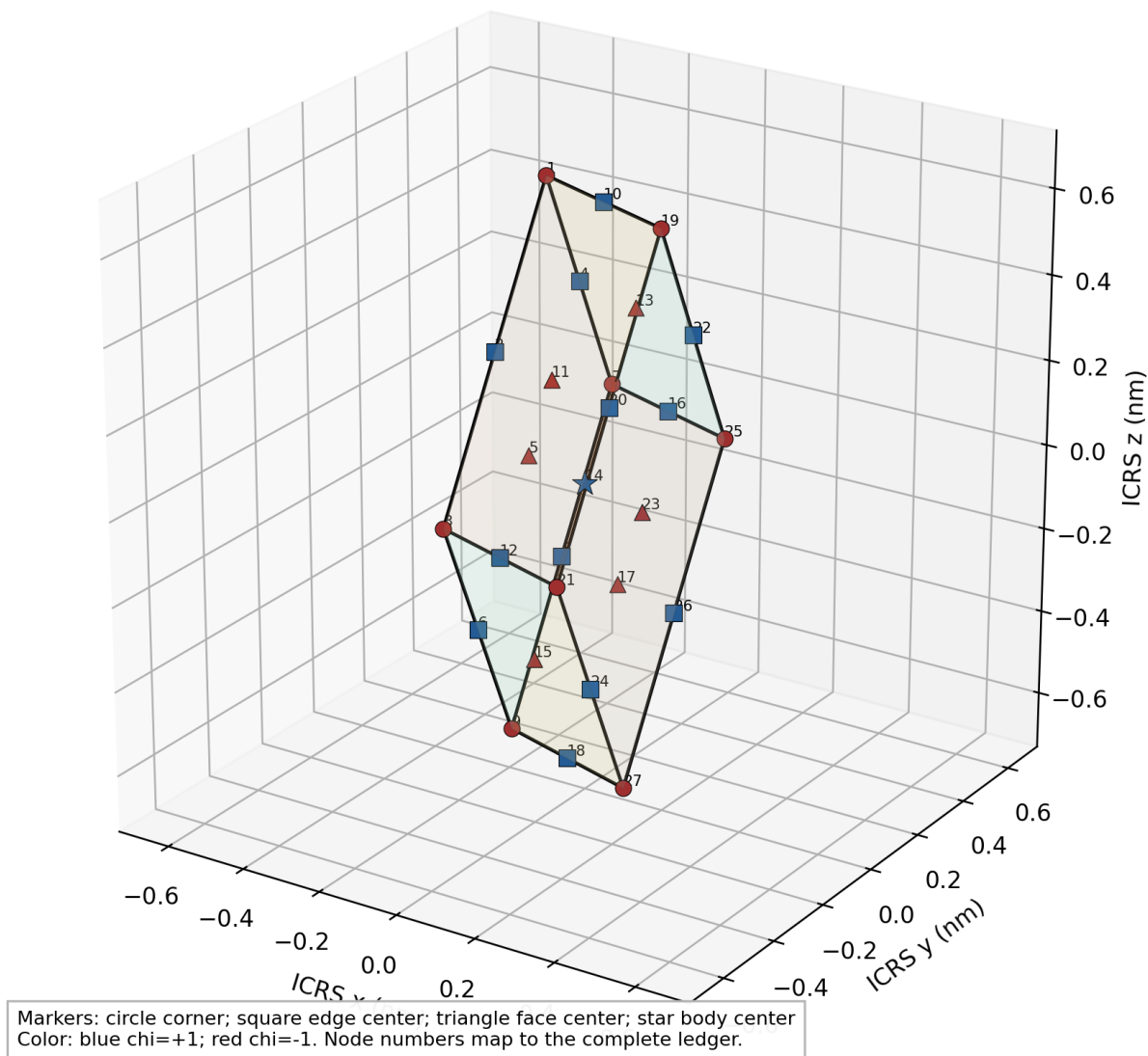


Figure 3. Dimensionally correct locked Cell3 in ICRS coordinates. Every centered node is numbered, classified, signed by polarity, and located within the six primitive boundary planes. The complete node ledger supplies the corresponding ternary address and coordinates.

Similarity freedom

$$\begin{aligned} B(s) &= s \ B_0 \\ M(s) &= s^2 \ M_0 \\ R(s) &= s^{-2} \ R_0 \\ A_{ij}(s) &= s^2 \ A_{ij,0} \\ V(s) &= s^3 \ V_0 \end{aligned}$$

Uniform scale changes every dimensional length, area, volume, and reciprocal wavenumber in the required way, but it cannot change edge ratios, angles, normalized metrics, plane-normal relations, projective face ratios, or recursive counts. The ontological content is the oriented affine similarity class together with its address, polarity, reciprocal, and recursive grammar.

4. The Mass Bridge Fixes the Cell3 Shape

The Mass Bridge is a locked projective ratio system. It is not a mechanical force that deforms a pre-existing cell. It is a constraint closure that identifies the one permitted Cell3 similarity shape within the admissible geometric construction.

Three primitive face areas form the geometric carrier triangle: $A_{ab}=|a \times b|$, $A_{ac}=|a \times c|$, and $A_{bc}=|b \times c|$. Their ratios carry the mass, fine-structure, and product channels. Because ratios are unchanged under uniform scaling, the Mass Bridge fixes shape independently of absolute size.

The face ratios are not isolated numerical similarities. They form a commutative triangle. Multiplying the ab-to-ac leg by the ac-to-bc leg cancels the intermediate area and yields the ab-to-bc leg. The corresponding physical channels multiply in the same order. This exact closure is the reason the three relations constitute one constraint system.

Mass-Bridge projective face-area closure

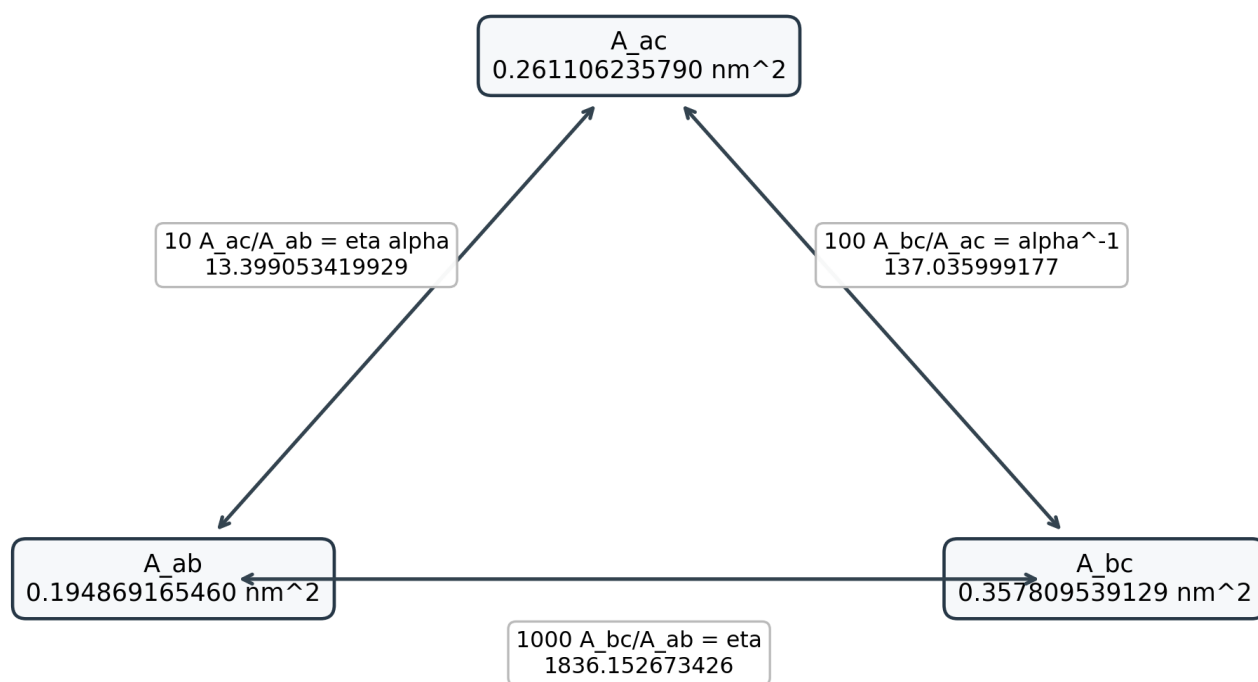


Figure 4. The Mass Bridge as a closed projective triangle. The three physical channels are ratios of the same three primitive face areas, and the product of the first two legs exactly equals the third.

Exact projective point

$$\begin{aligned} 1000 A_{bc}/A_{ab} &= \eta = 1836.152673426 \\ 100 A_{bc}/A_{ac} &= \alpha^{-1} = 137.035999177 \\ 10 A_{ac}/A_{ab} &= \eta \alpha = 13.399053419929222 \end{aligned}$$

$$(10 A_{ac}/A_{ab})(100 A_{bc}/A_{ac}) = 1000 A_{bc}/A_{ab}$$

The three relations fix the projective face-area point $[A_{bc}:A_{ac}:A_{ab}]$. Multiplying all three areas by one common positive factor preserves every relation. No independent angular or axial deformation remains after the complete locked construction is imposed. The only intrinsic continuous freedom is uniform scale.

5. Devon Dimension: The Reduced Algebraic Layer

The Devon Dimension is the compressed invariant representation of Cell3. It is the stage at which the local geometry has become algebraic but has not yet become a finite graph. Devon is not an additional spatial axis. It is the smallest typed algebraic package that preserves the direct cell, reciprocal response, face carriers, exchange hierarchy, scalar sockets, and rules required by later branches.

A compact Devon object is $D=(O_ICRS, B, M, \text{adj}(M), R, A, V, C, S, \text{Gamma2}, Q_rule, \text{branch sockets})$. Here A denotes the three primitive face areas, C denotes the carrier triangle, $S=R\text{-diag}(R)$ is the reciprocal exchange tensor, and Q_rule is $q(s)=s^T R s$. Typed branch sockets prevent rendered observables from being inserted upstream as geometric primitives.

The cofactor-dual identity is central. The diagonal cofactors of M are the squared areas of the opposite faces: $\text{adj}(M)_{aa}=A_{bc}^2$, $\text{adj}(M)_{bb}=A_{ac}^2$, and $\text{adj}(M)_{cc}=A_{ab}^2$. Since $R=\text{adj}(M)/V^2$, direct face geometry migrates exactly into reciprocal inverse-square geometry.

The off-diagonal reciprocal entries remain explicit. Their doubled magnitudes obey $|2R_{bc}| > |2R_{ac}| > |2R_{ab}|$. These exchange channels are already present in Devon. The Lucas Window does not create them; it supplies a finite address alphabet that evaluates and exposes them.

Devon Dimension: geometry compressed into invariant algebra

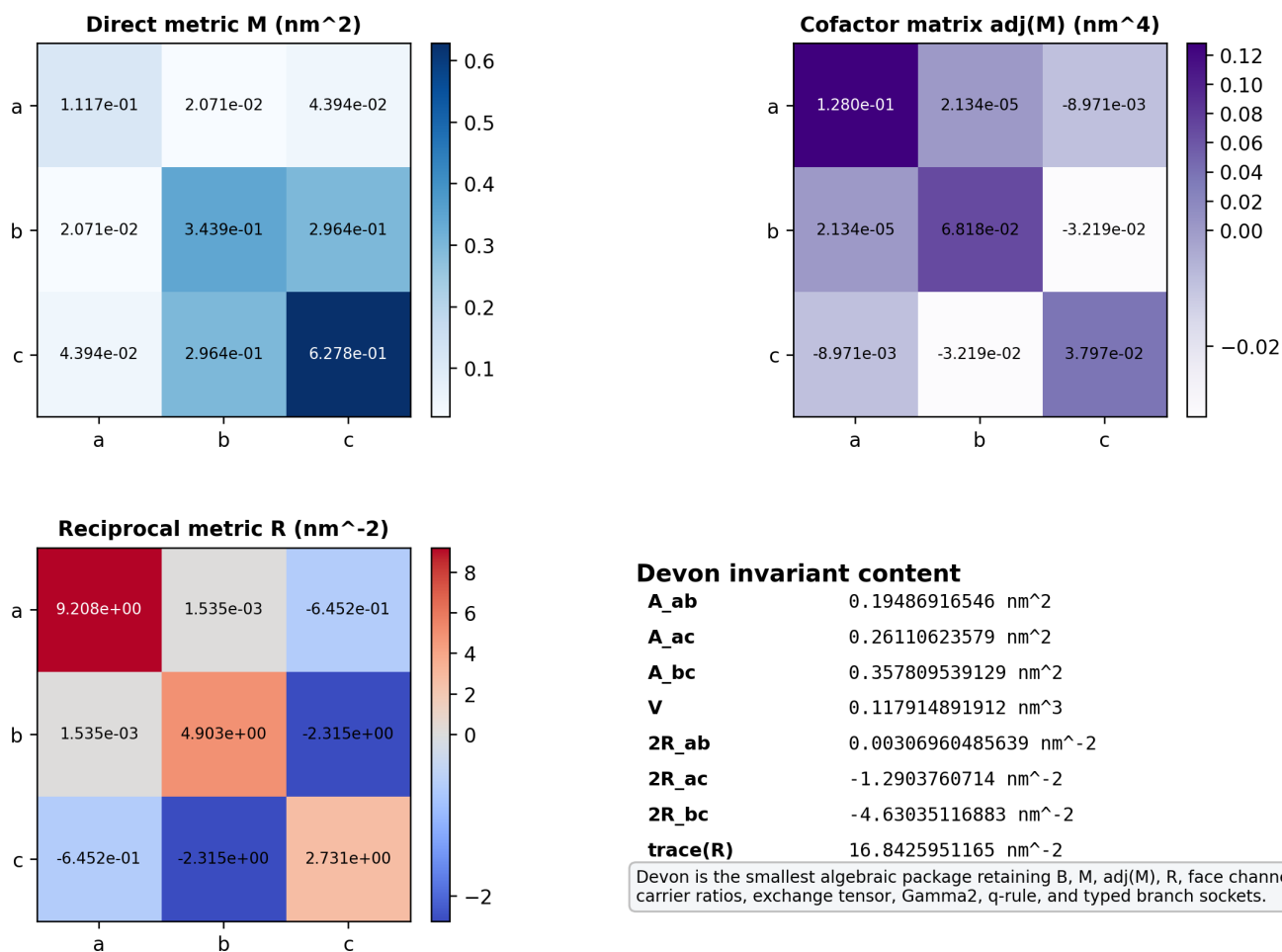


Figure 5. The Devon Dimension as the geometry-to-algebra reduction. The direct metric, cofactor matrix, reciprocal metric, face channels, volume, and exchange terms remain explicit; no finite graph has yet been imposed.

Devon content map

Object	Definition	Function
B	$[a\ b\ c]$	reconstructs the oriented direct cell
M	$B^T B$	lengths, angles, and direct quadratic geometry
$\text{adj}(M)$	cofactor matrix	face-to-reciprocal migration
R	M^{-1}	inverse-square geometry and q-spectrum weights
A_{ab}, A_{ac}, A_{bc}	primitive face areas	Mass-Bridge carrier channels
V	$ \det B $	normalizes cofactors into R
S	$R\text{-diag}(R)$	off-diagonal reciprocal exchange
Gamma2	native scalar socket	gravity-family branch input
Q_rule	$q(s) = s^T R s$	rule by which Lucas addresses interrogate Devon
branch sockets	typed interfaces	separate native structure from rendered observables

6. Direct Metric, Reciprocal Metric, and Plane Systems

The direct metric $M=B^T B$ converts lattice-coordinate displacements into physical squared lengths. If u is an address displacement, then $u^T M u$ is the squared physical distance represented by that displacement. The reciprocal metric $R=M^{-1}$ performs the dual operation for plane normals and reciprocal wavevectors.

The reciprocal basis B^{-T} contains three covectors g_a , g_b , and g_c satisfying $g_i \cdot a_j = \delta_{ij}$. A covector is a linear rule that assigns a scalar phase to a displacement. Each equation $g_i \cdot x = n$ defines one plane in an infinite family of parallel planes. Adjacent integer values of n identify adjacent planes.

Cell3 may therefore be viewed in two exact ways. In direct space it is the volume spanned by a , b , and c . In reciprocal space it is the region between adjacent members of three plane families. The planes are not an independent decorative overlay. They are the dual manifestation of the locked direct basis.

Direct-reciprocal duality of Cell3

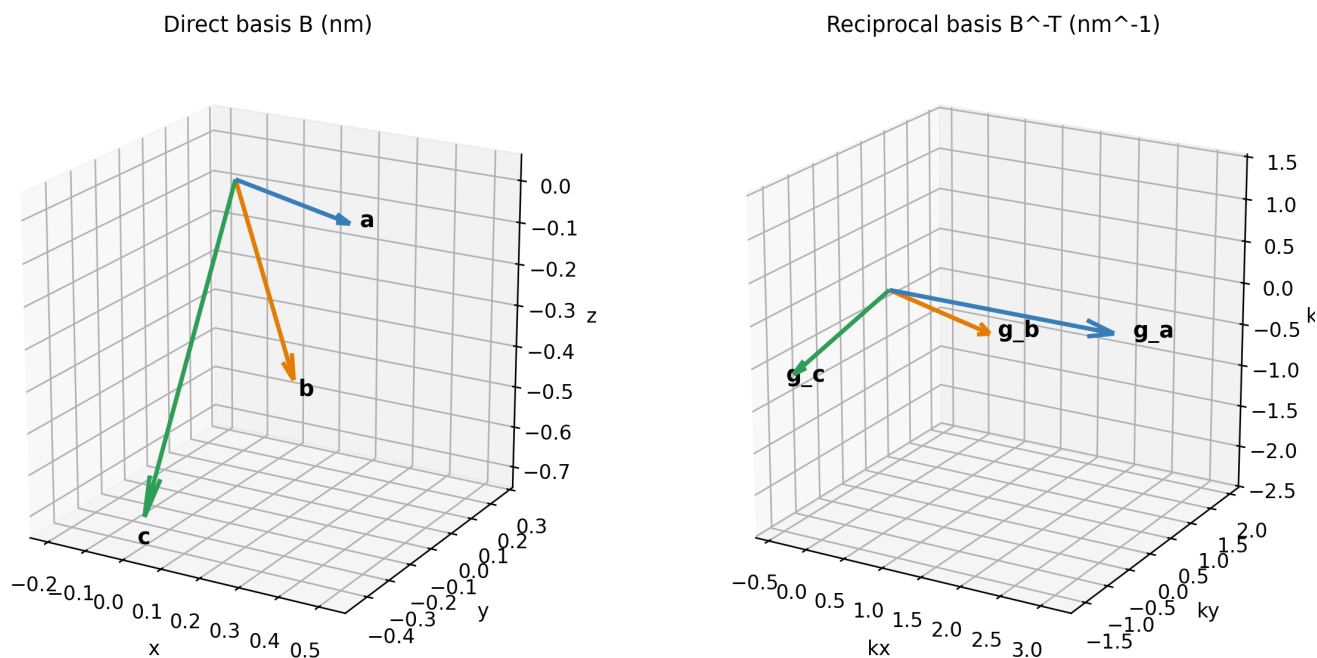


Figure 6. Direct basis vectors and their reciprocal dual. The relation $g_i \cdot a_j = \delta_{ij}$ makes each reciprocal vector normal to one primitive plane family and fixes its spacing.

Six primitive boundary planes in true in-plane metric

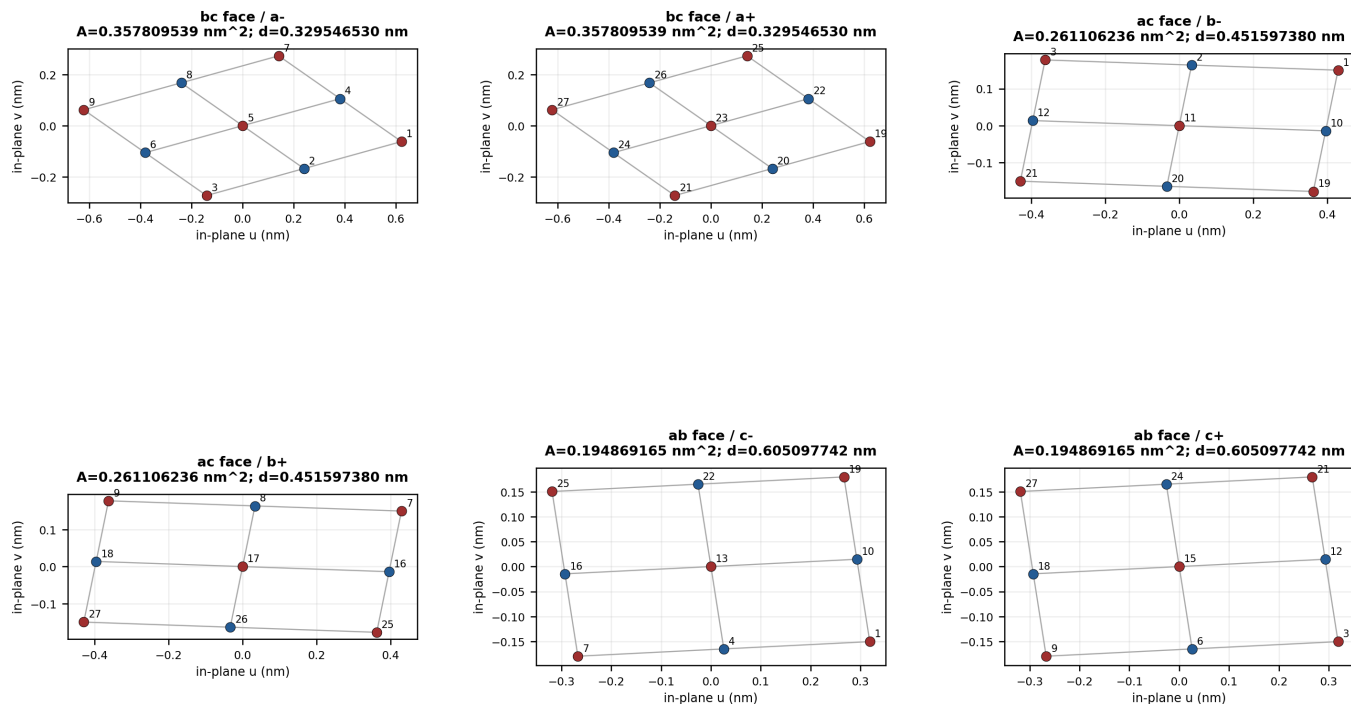


Figure 7. Six-plane atlas in true in-plane metric. Every panel contains all nine member nodes, numbered consistently with the complete Cell3 ledger, together with the exact face area and opposite-face spacing.

Cofactor and spacing relations

$$M = B^T B$$

$$R = M^{-1} = B^{-1} B^{-T}$$

$$\text{adj}(M)_{\text{diag}} = (A_{bc}^2, A_{ac}^2, A_{ab}^2)$$

$$d_a = V/A_{bc}$$

$$d_b = V/A_{ac}$$

$$d_c = V/A_{ab}$$

7. Fixed ICRS Orientation, Origin, and Observer State

The International Celestial Reference System supplies the coordinate frame in which the locked basis and face normals are recorded. ICRS labels are conventional, but the measured alignment relative to distant extragalactic directions is physical. Replacing right ascension and declination with another coordinate chart changes numerical components, not the underlying oriented tensors.

A fixed orientation produces daily and apparatus-controlled modulation. A physical affine origin adds translational phase. Rotation changes the direction sampled on the phase-cost sphere; translation changes the fractional Cell3 phase sampled in the lattice. These observables have different time signatures and must be modeled separately.

Default Observer-State Baseline and Directional Gravity Handling

Field	Authoritative default value
Frame	SGA-2020 / ICRS / J2000
Epoch	2026-06-17T15:06:11Z
Observer location	latitude 42.129 deg; longitude -83.144 deg; altitude 0 km
Observer speed	341.071130 km/s
Observer sky direction	RA 167.143423 deg; Dec -7.655565 deg
Reference / CMB closure	v_hat = (-0.97077086, 0.20734801, -0.12087493)
Angle from closure	1.066671 deg
Observer factors	f_obs = 1.447739; beta_obs^2 = 1.555741578671e-06
Directional vector	zeta_R = (0.320892, 0.374911, 0.838466)
Required G rule	Use the observer-indexed G-row chain or G_eff(n_hat) envelope. Do not pool scalar G rows with non-G scalar residual families unless direction, epoch, location, and compiler state are specified.

Observer-indexed gravity closure

$$\begin{aligned} \text{Gamma2} &= 2.400587821426810\text{e-}43 \\ \text{C_G_obs_SI} &= 2.780277372245049732\text{e}32 \\ \text{G_final} &= \text{Gamma2} \times \text{C_G_obs_SI} = 6.674300000000000\text{e-}11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2} \end{aligned}$$

The default packet therefore closes the selected scalar G row exactly at the CODATA witness value. Gamma2 remains the native geometric precursor; G_final is the observer-rendered SI quantity used by every downstream gravitational-coupling and Planck-family row.

8. Lucas Limit and Lucas Window

The Lucas Limit is the smallest finite extent that preserves the complete local three-axis structure. It is a minimum-completeness condition, not an asymptotic limit. Each primitive coordinate must contain a negative state, a central state, and a positive state. Their Cartesian product is the centered domain $L=\{-1,0,1\}^3$.

The Lucas Limit contains 27 addresses and 26 noncentral boundary states. A smaller domain cannot simultaneously preserve the center, opposing directions on all three axes, face, edge, and body address classes, antipodal pairing, polarity alternation, all 54 nearest-neighbor edges, and the complete seven-sheet partition.

The Lucas Window is the explicit 27-node graph realized at the Lucas Limit. Its abstract graph is $P3 \times P3 \times P3$. Two addresses are adjacent when they differ by one unit in exactly one coordinate. The physical embedding is $p(i,j,k)=(i \ a+j \ b+k \ c)/2$. The address grammar is rectilinear, but its dimensional realization is the oblique Cell3.

The 26 noncentral states form 13 antipodal pairs. Evaluating $q(s)=s^T R \ s$ on one representative of each pair gives 13 reciprocal magnitudes. Six values are sufficient to reconstruct a symmetric 3×3 metric; the remaining seven form exact compatibility or syndrome dimensions.

Lucas Limit and Lucas Window: minimum completeness and finite realization

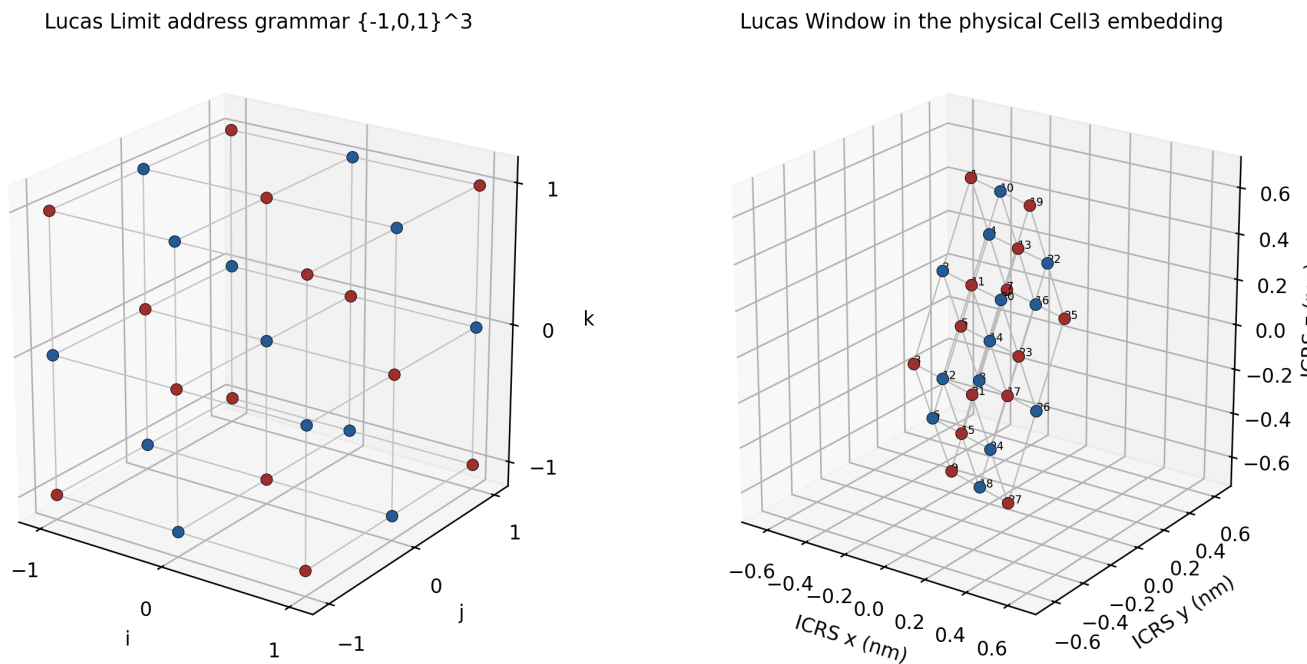


Figure 8. The Lucas Limit is the minimum-complete three-state address domain. The Lucas Window is its explicit 27-node graph placed by $p(i,j,k)=B(i,j,k)/2$ in the oblique Cell3 geometry.

Address classes

Class	Unsigned pattern	Signed states	Interpretation
Center	(0,0,0)	1	reference state
Face states	permutations of (1,0,0)	6	single-axis displacement
Edge states	permutations of (1,1,0)	12	two-axis displacement
Body states	(1,1,1)	8	three-axis displacement

9. The 27-State Centered Registry

The local Cell3 address set is $\{-1,0,1\}^3$. An address $L=(i,j,k)$ maps to physical position $p_L=B L/2$. The factor one-half places the outer addresses on the primitive boundaries, the zero coordinates on midplanes, and the origin at the body center.

The 27 states divide into eight corners, twelve edge centers, six face centers, and one body center. This is the coefficient structure of $(2+t)^3=8+12t+6t^2+t^3$. The classification records incidence dimension: a corner belongs to three boundary faces, an edge center belongs to two, a face center belongs to one, and the body center belongs to none.

Every noncentral state has one antipodal partner $L \leftrightarrow -L$. The 26 noncentral states therefore form 13 direction classes. Antipodal pairing removes sign duplication when a direction rather than an oriented ray is required, while retaining sign when polarity or directed motion is physically relevant.

Complete node ledger

ID	Address	Class	lambda	chi	x (nm)	y (nm)	z (nm)
1	(-1, -1, -1)	corner	-3	-1	-0.176351350136	0.118682429905	0.637311498325
2	(-1, -1, 0)	edge center	-2	1	-0.181416008950	-0.076211421278	0.292444294142
3	(-1, -1, 1)	corner	-1	-1	-0.186480667763	-0.271105272461	-0.052422910042
4	(-1, 0, -1)	edge center	-2	1	-0.158685962215	0.222342355125	0.363610955631
5	(-1, 0, 0)	face center	-1	-1	-0.163750621028	0.027448503942	0.018743751447
6	(-1, 0, 1)	edge center	0	1	-0.168815279841	-0.167445347241	-0.326123452736
7	(-1, 1, -1)	corner	-1	-1	-0.141020574293	0.326002280345	0.089910412937
8	(-1, 1, 0)	edge center	0	1	-0.146085233106	0.131108429162	-0.254956791247
9	(-1, 1, 1)	corner	1	-1	-0.151149891919	-0.063785422021	-0.599823995431
10	(0, -1, -1)	edge center	-2	1	-0.012600729109	0.091233925963	0.618567746878
11	(0, -1, 0)	face center	-1	-1	-0.017665387922	-0.103659925220	0.273700542694
12	(0, -1, 1)	edge center	0	1	-0.022730046735	-0.298553776403	-0.071166661489
13	(0, 0, -1)	face center	-1	-1	0.005064658813	0.194893851183	0.344867204184
14	(0, 0, 0)	body center	0	1	0.000000000000	0.000000000000	0.000000000000
15	(0, 0, 1)	face center	1	-1	-0.005064658813	-0.194893851183	-0.344867204184
16	(0, 1, -1)	edge center	0	1	0.022730046735	0.298553776403	0.071166661489
17	(0, 1, 0)	face center	1	-1	0.017665387922	0.103659925220	-0.273700542694
18	(0, 1, 1)	edge center	2	1	0.012600729109	-0.091233925963	-0.618567746878
19	(1, -1, -1)	corner	-1	-1	0.151149891919	0.063785422021	0.599823995431
20	(1, -1, 0)	edge center	0	1	0.146085233106	-0.131108429162	0.254956791247
21	(1, -1, 1)	corner	1	-1	0.141020574293	-0.326002280345	-0.089910412937
22	(1, 0, -1)	edge center	0	1	0.168815279841	0.167445347241	0.326123452736
23	(1, 0, 0)	face center	1	-1	0.163750621028	-0.027448503942	-0.018743751447
24	(1, 0, 1)	edge center	2	1	0.158685962215	-0.222342355125	-0.363610955631
25	(1, 1, -1)	corner	1	-1	0.186480667763	0.271105272461	0.052422910042
26	(1, 1, 0)	edge center	2	1	0.181416008950	0.076211421278	-0.292444294142
27	(1, 1, 1)	corner	3	-1	0.176351350136	-0.118682429905	-0.637311498325

10. Lucas Sheets and Devon Super Sheets

The laminar grade is $\lambda = i + j + k$. Because each coordinate is -1, 0, or 1, λ ranges from -3 to +3. States with equal λ lie on one plane normal to the common laminar direction. Their populations are 1, 3, 6, 7, 6, 3, and 1.

The polarity is $\chi = (-1)^\lambda$. Every state on one sheet therefore has the same polarity, and adjacent sheets have opposite polarity.

Displacements remaining within one sheet satisfy $\Delta i + \Delta j + \Delta k = 0$. They form an A2-type planar kernel embedded in the three-dimensional oblique cell. The sheet system therefore combines three-dimensional volume with two-dimensional transport and incidence structure.

Lucas Sheet Decomposition of the minimum-complete window

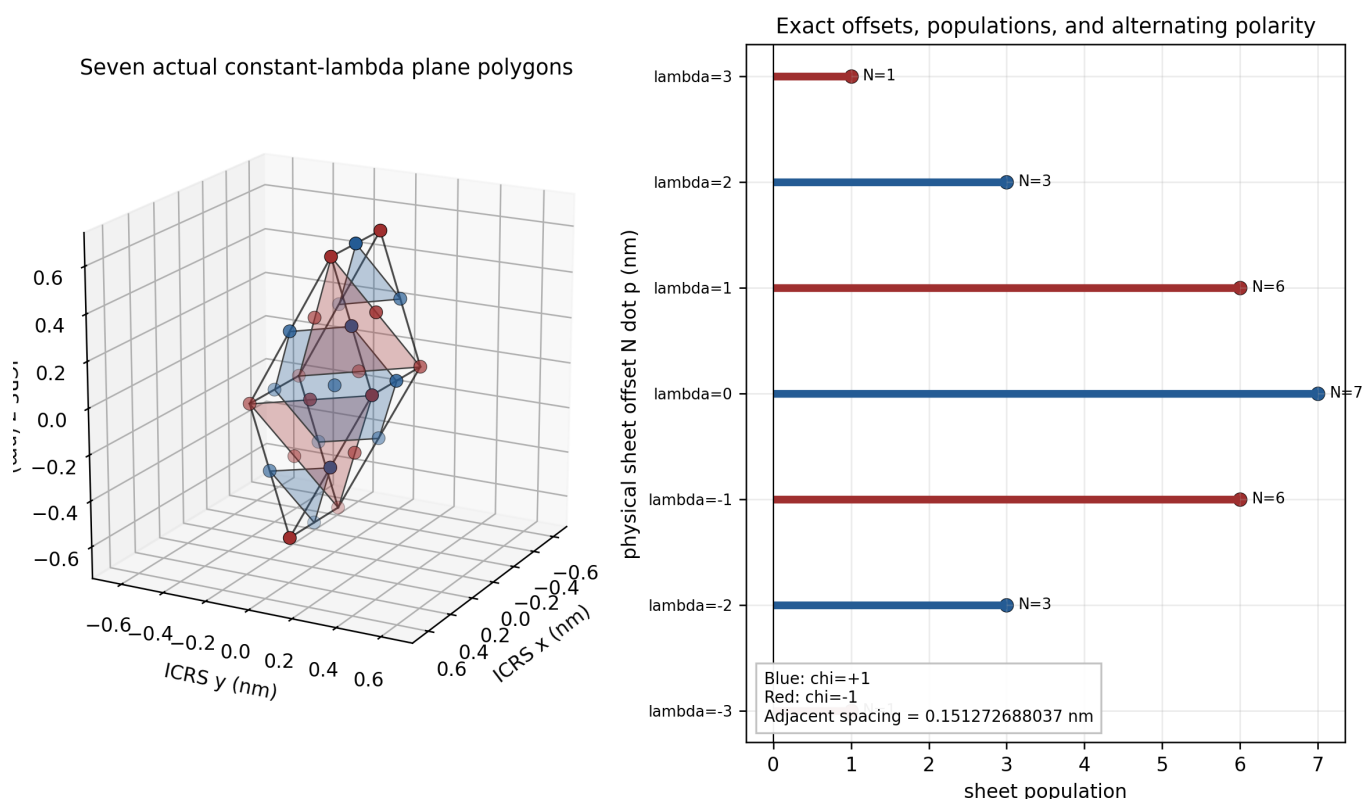


Figure 9. The seven Lucas Sheets are actual parallel planes in the oblique physical embedding, not only a population sequence. Their grades are $\lambda = -3$ through $+3$, their populations are 1, 3, 6, 7, 6, 3, 1, and adjacent sheets alternate polarity.

From seven local sheets to fifty-five super sheets

At Lucas-Supergroup scale the refined coordinates range from -9 through +9 on each primitive axis. The laminar grade therefore spans $\lambda = -27$ through $+27$, producing 55 actual parallel planes. These larger constant-lambda planes are Devon Super Sheets. They preserve the same polarity law $\chi = (-1)^\lambda$ and the same physical normal as the seven local Lucas Sheets.

Seven carrier positions occur at $\lambda = -27, -18, -9, 0, 9, 18, 27$. The exact sheet profile sums to 6,859 refined states. The positive and negative populations differ by exactly one, so the net polarity of every exact Lucas closure is -1.

Devon Super Sheets in the 9B Lucas Supergroup

All 55 plane outlines; seven carrier sheets filled

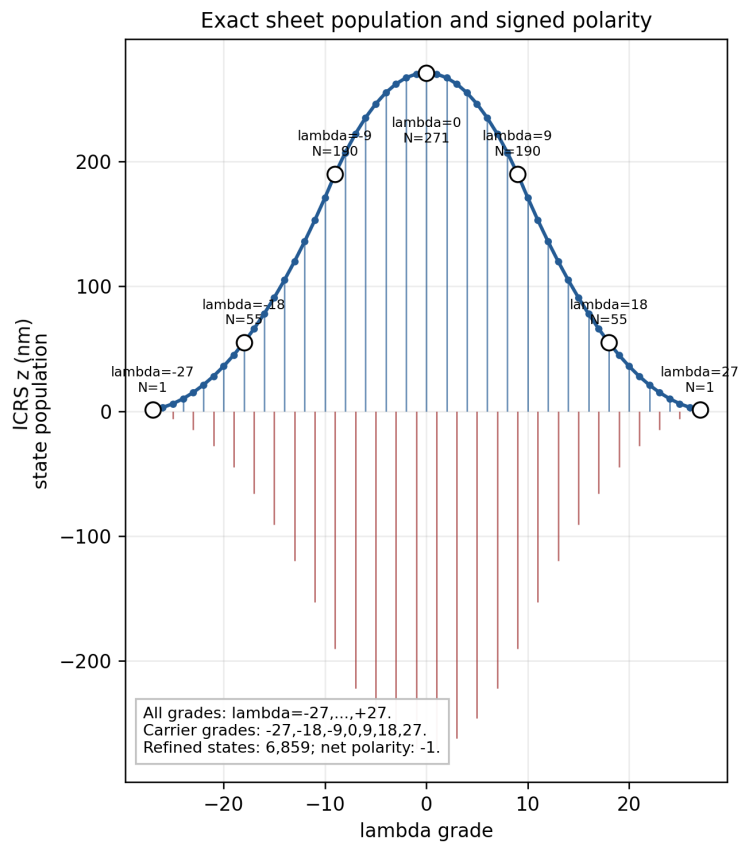
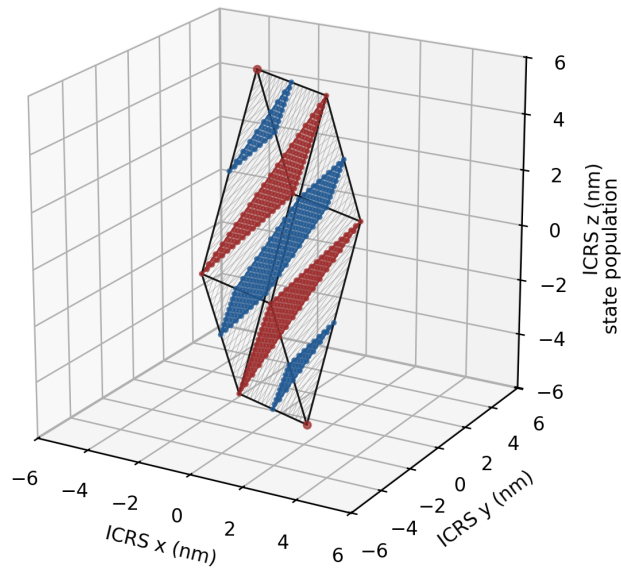


Figure 10. The complete 55-sheet geometry of the 9B Lucas Supergroup. Every sheet is the physical plane $\lambda = i+j+k$ in the 19-by-19-by-19 refined lattice. Seven harmonic carrier sheets are emphasized, while the population profile and alternating polarity are shown from the exact 6,859-state census.

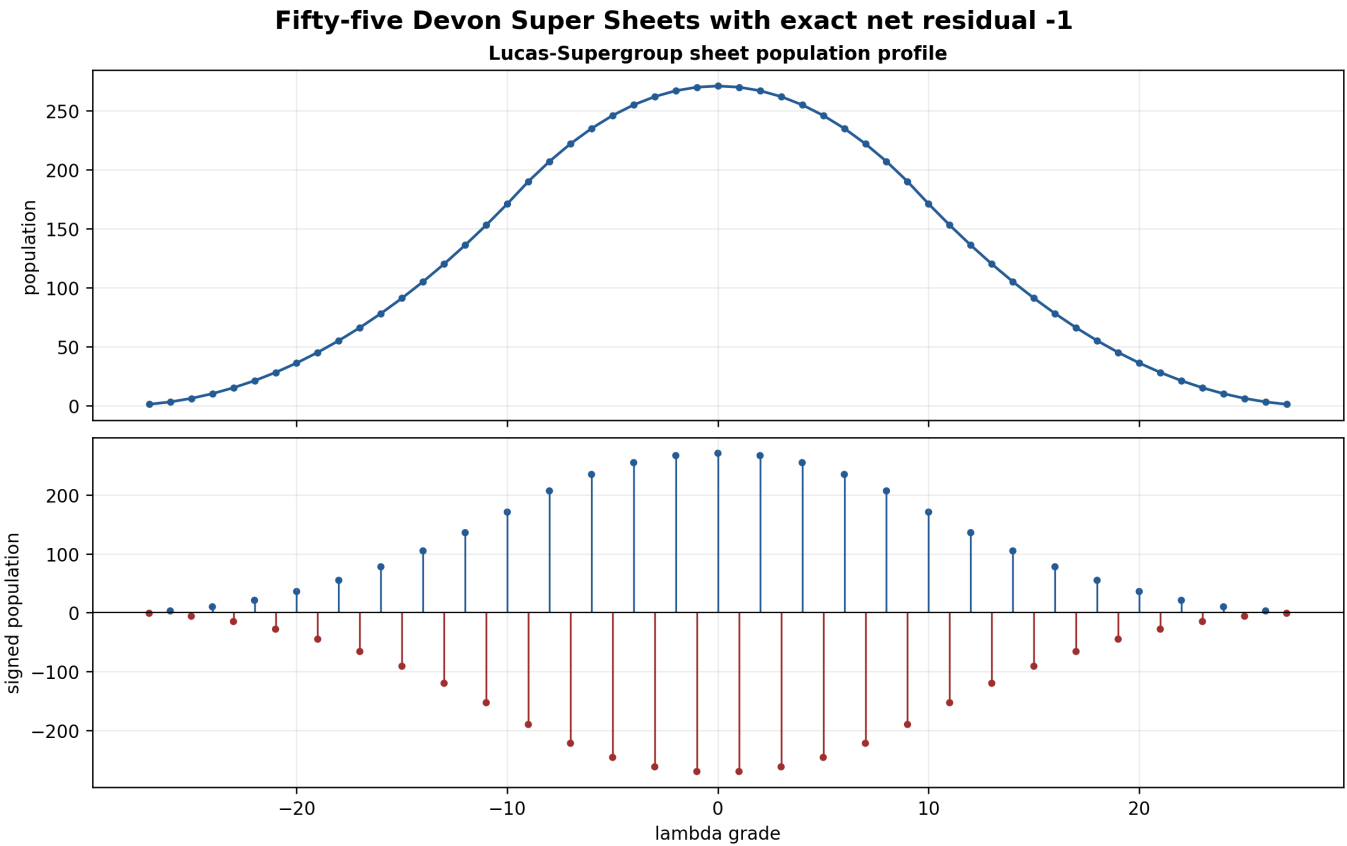


Figure 11. The exact 55-sheet population and signed-polarity profile for the Lucas Supergroup. Positive and negative populations grow extensively, but their global difference remains exactly one state.

11. Emily Eigen-Spectrum

The Emily Eigen-Spectrum is the adjacency spectrum of the finite Lucas graph $P_3 \times P_3 \times P_3$. It is a spectral reading of connectivity and is independent of the absolute Cell3 scale. The physical basis determines where the nodes are placed; the adjacency operator determines how one-step relations propagate through the finite window.

The one-dimensional path P_3 has adjacency eigenvalues $-\sqrt{2}$, 0 , and $+\sqrt{2}$. The three-dimensional Lucas adjacency operator is the Kronecker sum $A_L = A_{P_3} \otimes I \otimes I + I \otimes A_{P_3} \otimes I + I \otimes I \otimes A_{P_3}$. A tensor-product mode has eigenvalue equal to the sum of its three one-axis eigenvalues.

Grouping equal sums gives seven eigenvalue levels: $-3\sqrt{2}$, $-2\sqrt{2}$, $-\sqrt{2}$, 0 , $\sqrt{2}$, $2\sqrt{2}$, and $3\sqrt{2}$. Their multiplicities are 1,3,6,7,6,3,1. The adjacency nullity is seven; the graph-Laplacian nullity is one because the graph is connected.

The central multiplicity seven has a second structural interpretation. Thirteen antipodal q measurements reconstruct six independent metric coefficients and leave seven compatibility dimensions. Emily therefore exposes in spectral form the same seven-dimensional surplus that appears in the projective metric-syndrome decomposition.

Emily: intrinsic modes of the minimum-complete Lucas graph

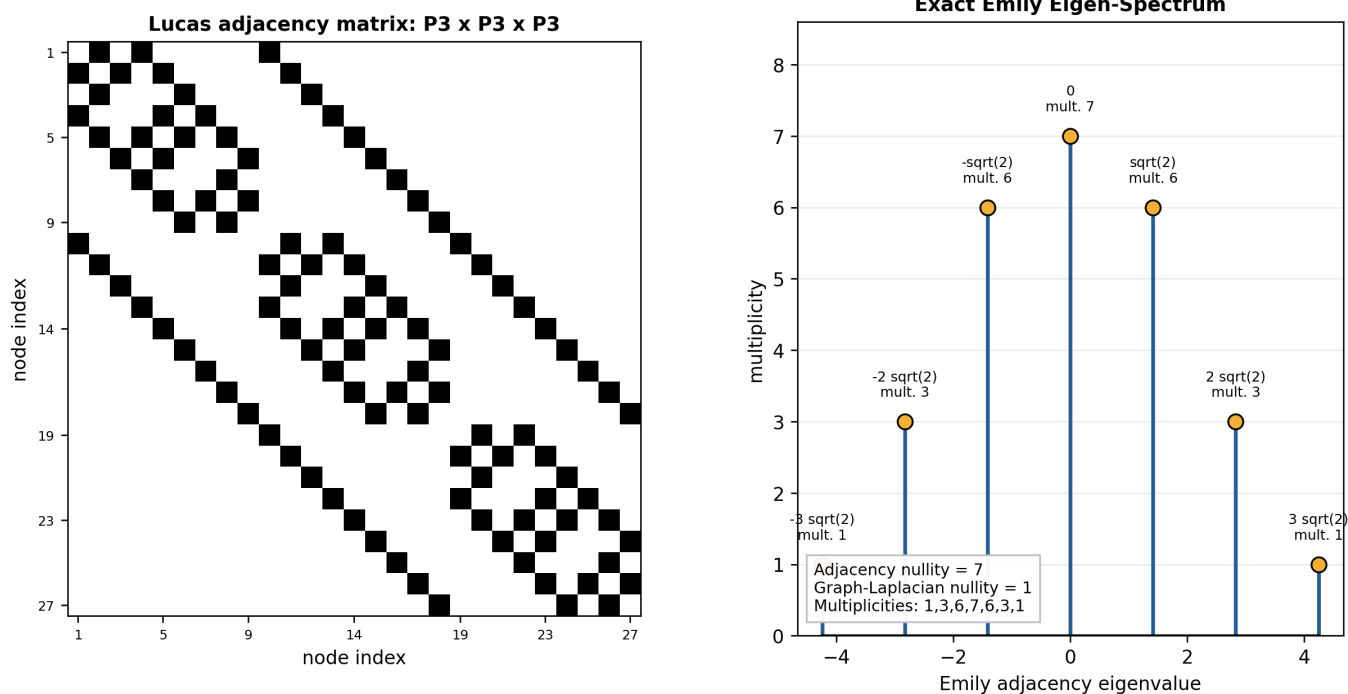


Figure 12. Emily is the adjacency spectrum of $P_3 \times P_3 \times P_3$. The seven eigenvalue levels have multiplicities 1,3,6,7,6,3,1; the central zero eigenspace has dimension seven, matching the seven compatibility dimensions left after thirteen projective q measurements reconstruct six metric coefficients.

Eigenvalue	Numerical value	Multiplicity
$-3\sqrt{2}$	-4.242640687119286	1
$-2\sqrt{2}$	-2.828427124746190	3
$-\sqrt{2}$	-1.414213562373095	6
0	0.000000000000000	7
$\sqrt{2}$	1.414213562373095	6
$2\sqrt{2}$	2.828427124746190	3
$3\sqrt{2}$	4.242640687119286	1

12. Jack Jacobian

The Jack Jacobian is the differential response map that carries coherent primitive-axis perturbations into changes of selected Devon invariants. It is not a new force law and not another cell. It is the linearized derivative grammar of the reduced hierarchy.

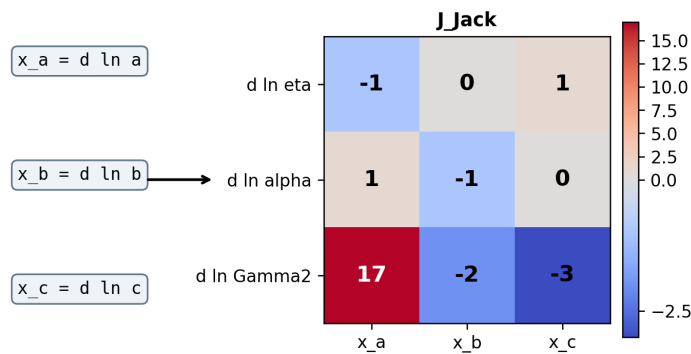
Let $x_a = d \ln a$, $x_b = d \ln b$, and $x_c = d \ln c$ denote coherent logarithmic perturbations of the three primitive directions. The invariant responses are $d \ln \eta = -x_a + x_c$, $d \ln \alpha = x_a - x_b$, and $d \ln \Gamma_2 = 17x_a - 2x_b - 3x_c$.

The coefficient matrix is $J_{\text{Jack}} = \begin{bmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 17 & -2 & -3 \end{bmatrix}$. Its determinant is 12, so the map is nondegenerate and locally invertible. Its Smith normal form is $\text{diag}(1, 1, 12)$, giving the finite quotient Z_{12} .

The inverse relations reconstruct the coherent primitive perturbations from the invariant changes. Jack can therefore propagate a Cell3 deformation into Devon, Lucas q-values, metric-weighted spectra, and downstream rendered branches, or work backward from a measured invariant perturbation to the local primitive-axis response.

Jack Jacobian: differential response across Cell3, Devon, and downstream branches

Primitive coherent perturbations



Local inverse and algebraic index

$$x_a = (dG - 2 dA + 3 dE) / 12$$

$$x_b = (dG - 14 dA + 3 dE) / 12$$

$$x_c = (dG - 2 dA + 15 dE) / 12$$

$$\det(J_{\text{Jack}}) = 12$$

$$\text{Smith normal form} = \text{diag}(1, 1, 12)$$

$$\text{cokernel quotient} = Z_{12}$$

$dE = d \ln \eta$; $dA = d \ln \alpha$; $dG = d \ln \Gamma_2$. Nonzero determinant makes the forward sensitivity map locally invertible.

Figure 13. The Jack Jacobian maps coherent logarithmic changes of the three primitive directions into changes of eta, alpha, and Gamma2. Its determinant is 12, its Smith normal form is $\text{diag}(1, 1, 12)$, and its finite quotient is Z_{12} .

$$J_{\text{Jack}} = \begin{bmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 17 & -2 & -3 \end{bmatrix}$$

$$\det(J_{\text{Jack}}) = 12$$

$$\text{SNF}(J_{\text{Jack}}) = \text{diag}(1, 1, 12)$$

$$\text{cokernel} = Z_{12}$$

Recursive Lucas-Jack index

The determinant 12 also appears as $27(2/3)^2 = 12$ when the 27-state Lucas count is combined with the quadratic depth factor associated with the native $3/2$ recursion. This equality links the finite graph branching count to the differential exposure index without identifying the two objects as the same mathematical type.

13. Quadratic Phase-Transition Cost

The K-Field assigns a direction-dependent transition cost through the positive-definite quadratic form $q(n)=n^T G_{\text{phase}} n$ on the unit sphere. The value depends on direction, not on the coordinate labels used to represent that direction.

The three eigenvectors of G_{phase} define the soft, saddle, and hard axes. The soft axis minimizes q , the hard axis maximizes q , and the saddle axis is intermediate. The current eigenvalues are 1.4293287534874834, 2.1125450264474788, and 8.800336434665038.

The hard-to-soft anisotropy ratio is approximately 6.157. An otherwise equivalent directed transition can therefore sample substantially different phase cost when its orientation changes relative to the locked K-Field tensor.

Constant- q contours are intersections between the unit sphere and a family of concentric quadratic surfaces. Below the middle eigenvalue, the contours encircle the soft axis. Above it, complementary contours organize around the hard axis. The middle eigenvalue defines the separatrix topology associated with the saddle axis.

Quadratic phase-transition cost $q(n)=n^T G_{\text{phase}} n$

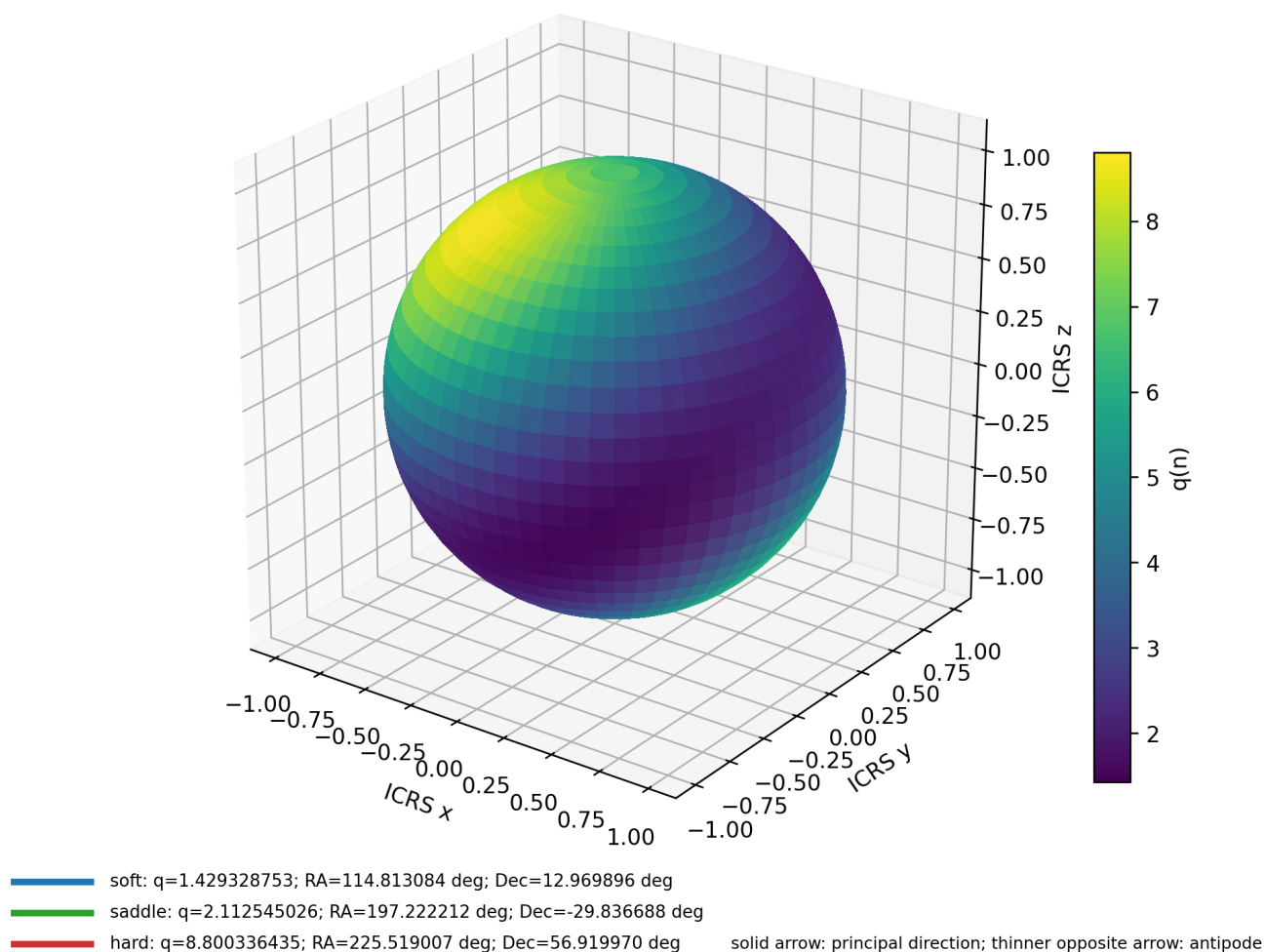


Figure 14. The quadratic phase-transition cost on the ICRS unit sphere. Soft, saddle, and hard axes are the eigenvectors of G_{phase} ; their eigenvalues define the minimum, intermediate, and maximum directional costs.

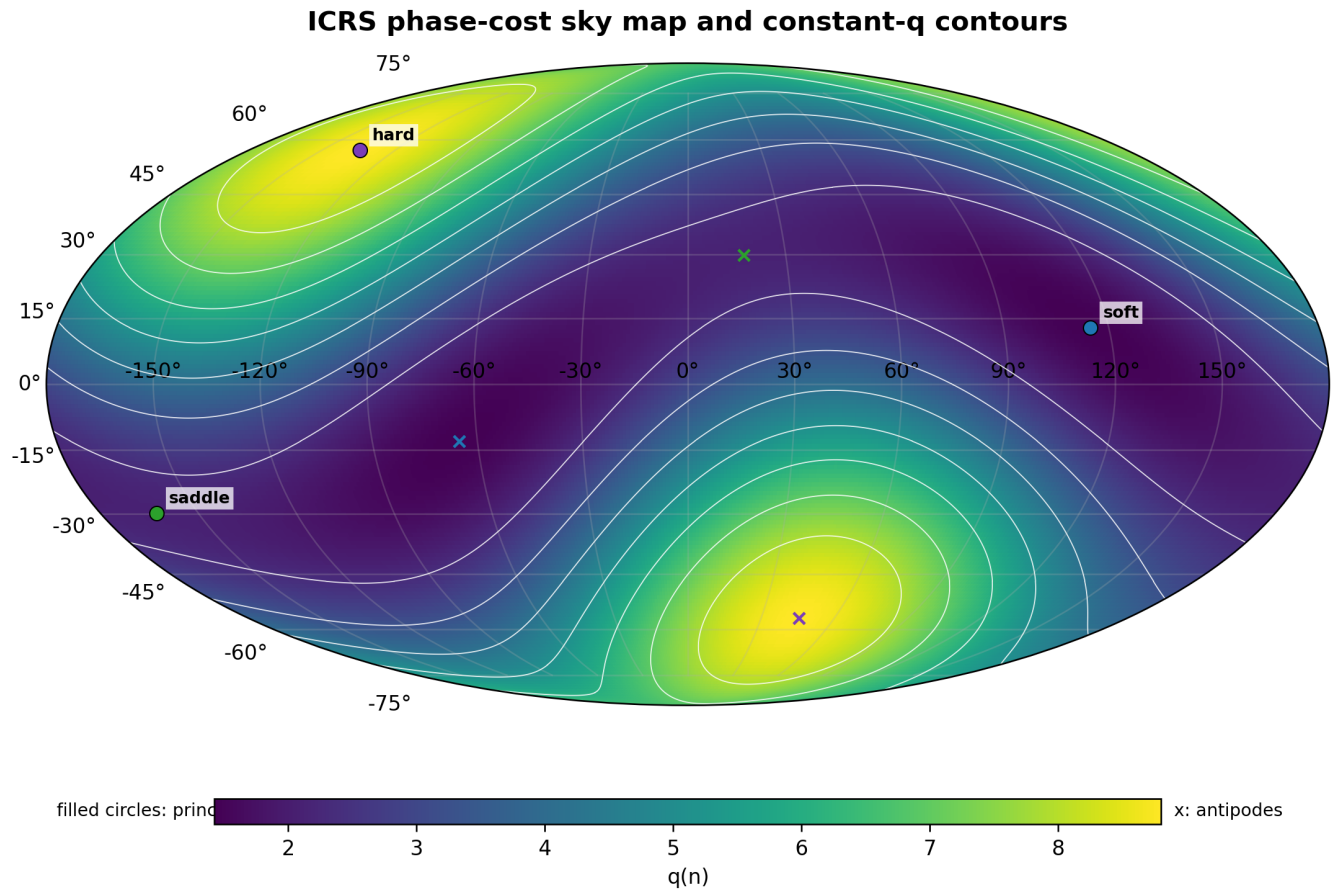


Figure 15. Mollweide projection of the exact phase-cost function in ICRS coordinates. White curves are constant-cost contours; labeled points are the principal axes and x marks their antipodes. The legend and color scale occupy a dedicated non-overlapping band.

Cost-conserving orientation flow

$$dn/dt = -(2/L) \, n \times (G_{\text{phase}} \, n)$$

$$d(n \cdot n)/dt = 0$$

$$dq/dt = 0$$

$$|dn/dt| = (2/L) |n \times G_{\text{phase}} \, n|$$

The flow remains on the unit sphere and on one constant- q contour. The principal axes are stationary because $G_{\text{phase}} \, n$ is parallel to n at an eigenvector. Residence time therefore increases near orientations where the angular speed becomes small.

14. Metastability, Path Multiplicity, and Entropy

The K-Field geometry is deterministic, but metastable outcomes may remain probabilistic. A deterministic state structure specifies the available endpoints and transitions. Probability enters because many distinct ordered histories can terminate at the same endpoint.

Let $\Omega_N(k)$ be the number of N -event histories reaching endpoint k . The equal-weight fiber entropy is $\ln \Omega_N(k)$. A high-multiplicity endpoint collects more histories than a low-multiplicity endpoint even when their geometric displacement is similar.

For the exact ten-event enumeration, $6^{10}=60,466,176$ ordered histories terminate on 891 reachable endpoints. The history entropy is 17.91759469228055 nats. The endpoint entropy is 5.364728188272016 nats. The mean conditional fiber entropy is 12.552866504008534 nats, and the chain-rule residual is approximately -1.78×10^{-15} nats.

The combined endpoint weight contains both multiplicity and cost: $P_N(k)$ proportional to $\Omega_N(k) \exp[-\beta_K q(k)]$. Low cost favors a state energetically or geometrically; high multiplicity favors it statistically. The effective action $\beta_K q - \ln \Omega$ expresses the competition in one scalar.

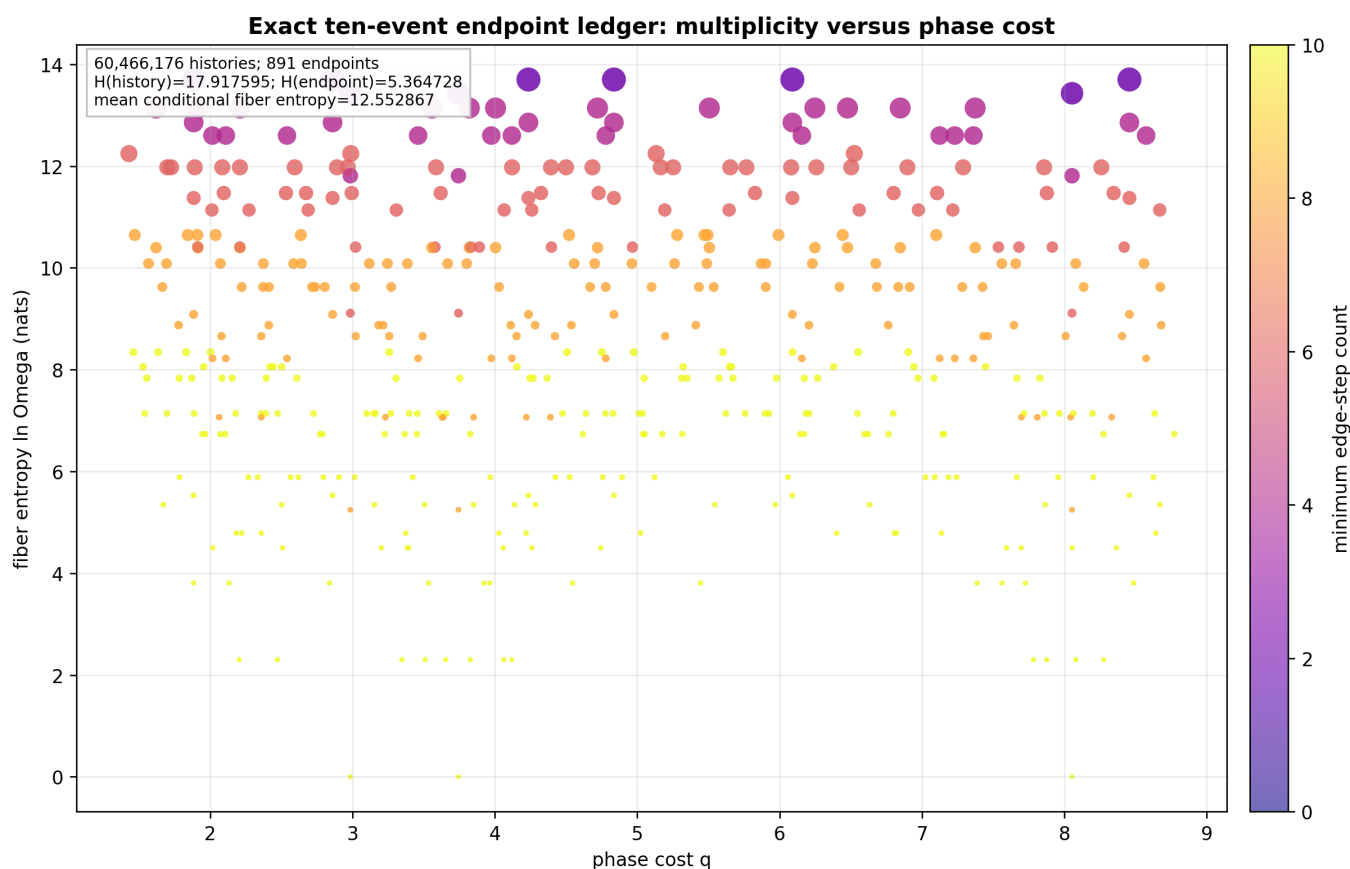


Figure 16. Exact ten-event enumeration. Each point is a reachable endpoint; height is path-fiber entropy, horizontal position is phase cost, color is minimum edge-step distance, and marker area scales with endpoint probability.

Unified population measure

$$P_N(n) \propto \Omega_N(n) \exp[-\beta_K q(n)] C(n) / |n \times G_{\text{phase}} n|$$

The factors represent path multiplicity, phase cost, curvature organization, and precessional residence time. They are separate contributions and must not be collapsed into a single undifferentiated scalar.

15. Scalar Serialization and Nonlocal Address Order

The three-dimensional K-Field lattice admits a lossless one-dimensional directed address. A positive quadratic shell $Q(k)=k^T G k$ is combined with an antipodal sign $\sigma(k)$ to form $\rho(k)=\sigma(k)\sqrt{Q(k)}$. Under the recorded shell condition, ρ is injective: equal scalar addresses imply equal lattice addresses.

Injectivity makes ρ a complete address coordinate, but it does not make ρ a physical distance. The ordering can place physically adjacent nodes far apart in scalar rank. Conversely, two consecutive scalar ranks need not be geometric nearest neighbors.

The exact 27-state comb contains 54 geometric nearest-neighbor edges. Only eight join consecutive scalar ranks. The mean rank jump is 5.8148148148, the median is 4, and the maximum is 23. Mean jumps along the primitive a, b, and c directions are 7.3333, 4.1111, and 6.0000.

Two graphs must therefore be superposed. G_{geo} records physical adjacency. G_{ρ} records scalar ordering. A short edge in G_{geo} that spans a long interval in G_{ρ} is an exact serialized-space shortcut. Such edges are natural carriers for routing, error correction, history encoding, and nonlocal address transitions.

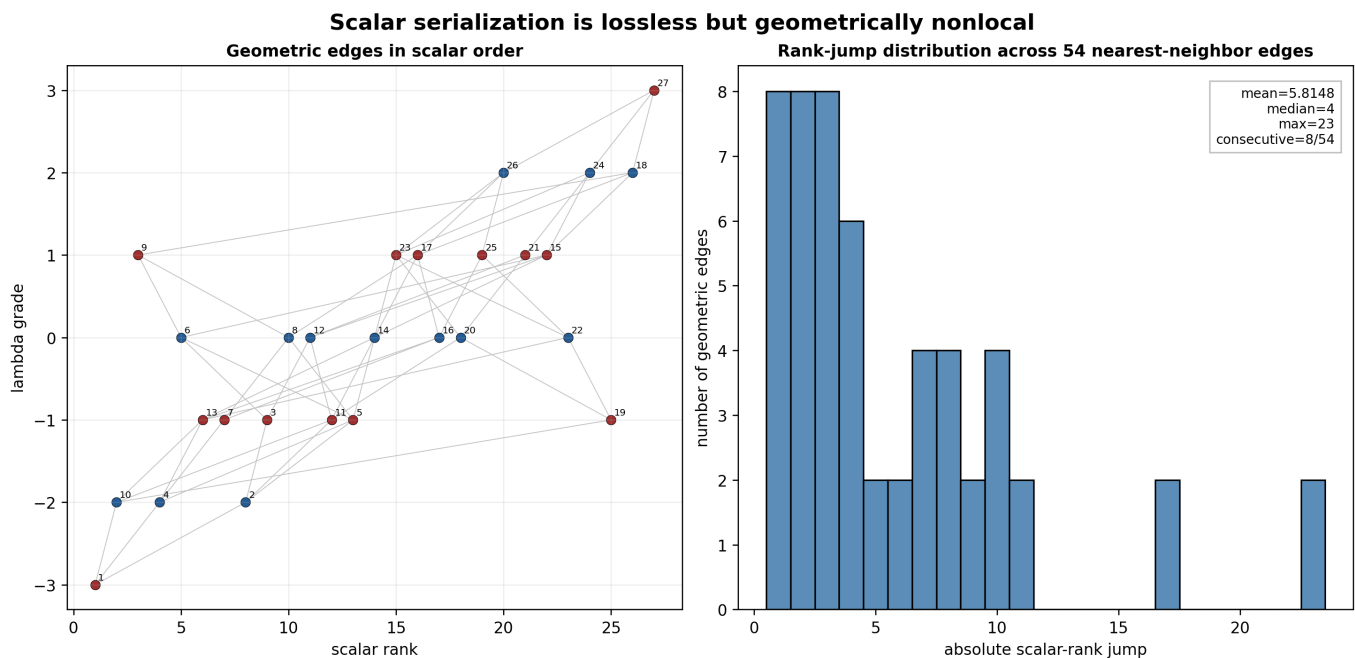


Figure 17. The exact 27-state scalar comb and the scalar-rank jumps of all 54 geometric nearest-neighbor edges. Only eight edges connect consecutive scalar ranks, so ρ is a directed address rather than a locality coordinate.

16. Thirteen Projective Channels

The 26 noncentral Cell3 addresses become 13 antipodal direction classes when L and -L are identified. These classes form the finite projective plane PG(2,3). It contains 13 projective points, 13 projective lines, and 52 point-line incidences.

The incidence matrix N satisfies $N^2=3I+J$. Its eigenvalues are 4 and six copies each of $+\sqrt{3}$ and $-\sqrt{3}$. The associated transform $M=3N-J$ satisfies $M^2=27I-2J$. These identities encode exact symmetry and inversion properties of the channel system.

The 13 channels separate into six metric channels and seven syndrome channels. A symmetric 3 x 3 metric has six independent coefficients. The six positive channels provide a full-rank decoder for those coefficients. The remaining seven channels test compatibility and retain information not required by the metric reconstruction itself.

This 6+7 decomposition links geometry and information. Metric channels estimate the quadratic form. Syndrome channels identify deviations, errors, path-history effects, or additional state information. The same 13-channel architecture therefore supports both measurement and consistency testing.

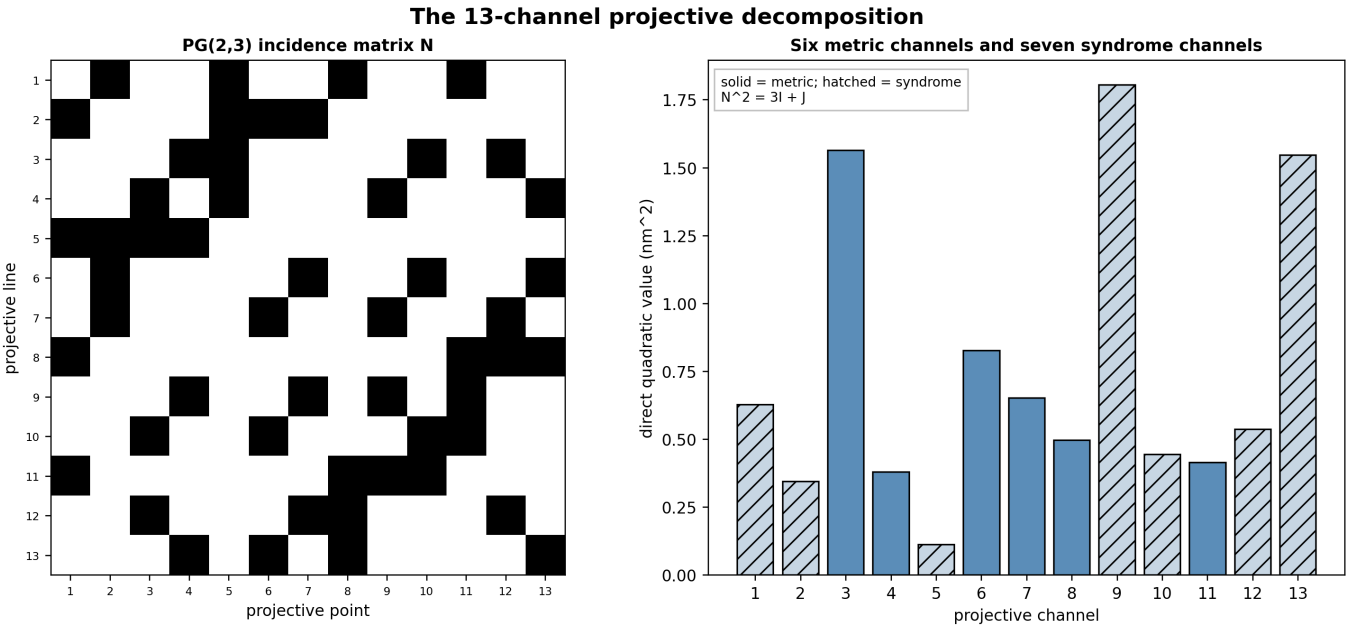


Figure 18. The finite projective plane PG(2,3) and the 6+7 channel decomposition. Six metric channels reconstruct a symmetric three-dimensional quadratic form; seven syndrome channels supply compatibility and path-sensitive residual dimensions.

ID	Representative	Role	chi	q_direct (nm ²)	q_recip (nm ⁻²)	Incidences
1	(0, 0, 1)	syndrome	-1	0.627770610077	2.731171274712	2,5,8,11
2	(0, 1, 0)	syndrome	-1	0.343877732393	4.903398312368	1,5,6,7
3	(0, 1, 1)	metric	1	1.564433860393	3.004218418247	4,5,10,12
4	(0, 1, -1)	metric	1	0.378862824547	12.264920755912	3,5,9,13
5	(1, 0, 0)	syndrome	-1	0.111676057896	9.208025529458	1,2,3,4
6	(1, 0, 1)	metric	1	0.827321138166	10.648820732769	2,7,10,13
7	(1, 0, -1)	metric	1	0.651572197781	13.229572875569	2,6,9,12
8	(1, 1, 0)	metric	1	0.496974456850	14.114493446682	1,11,12,13
9	(1, 1, 1)	syndrome	-1	1.805405055043	10.924937481161	4,7,9,11
10	(1, 1, -1)	syndrome	-1	0.444085078812	22.766391961626	3,6,10,11
11	(1, -1, 0)	metric	1	0.414133123728	14.108354236969	1,8,9,10
12	(1, -1, 1)	syndrome	-1	0.536992686074	20.179500609114	3,7,8,12
13	(1, -1, -1)	syndrome	-1	1.546814781536	13.499550414248	4,6,8,13

17. Lucas Groups and the Lucas Supergroup

Cell3 fills volume through powers of three. A Lucas Group is a $3 \times 3 \times 3$ array containing 27 complete Cell3 volumes. Its outer basis is 3B. A Lucas Supergroup is a $9 \times 9 \times 9$ array containing 729 Cell3 volumes, or equivalently 27 Lucas Groups. Its outer basis is 9B.

The Lucas Supergroup has cubical census $(f_0, f_1, f_2, f_3) = (1000, 2700, 2430, 729)$. It contains 6,859 refined half-step states, 55 laminar sheets, 1,944 internal interfaces, 1,946 boundary states, 486 exposed primitive faces, and center-graph cycle rank 1,216.

The complete cubical complex is contractible even though its center graph contains many cycles. Graph cycles become boundaries when the intervening plaquettes are included. Local circulation and global topological closure therefore coexist without contradiction.

The Mass Bridge selects the second Lucas aggregation. Nine cells along one axis have ten unrefined boundary-node positions. The one-, two-, and three-dimensional cardinalities are 10, 100, and 1000, exactly matching the projective multipliers in the Mass Bridge.

Exact volume-filling aggregation: Cell3, Lucas Group, Lucas Supergroup

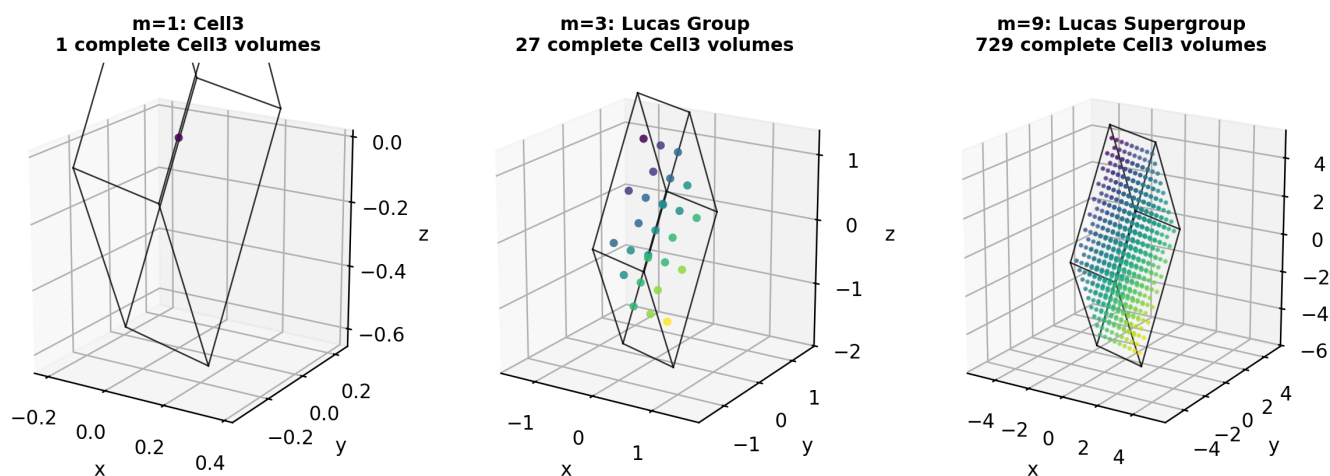


Figure 19. Exact spatial aggregation at $m=1, 3, 9$. The basis scales as mB while the Cell3 shape is preserved. The Lucas Group contains 27 cells; the Lucas Supergroup contains 729.

Lucas-Supergroup exact census

Invariant	Exact value
outer basis	9B
Cell3 volumes	729
Lucas Groups	27
f-vector	(1000, 2700, 2430, 729)
refined states	6,859
laminar sheets	55
central sheet population	271
internal interfaces	1,944
boundary states	1,946
exposed primitive faces	486
center-graph cycle rank	1,216

18. Exact Recursive Morphology

At aggregation depth g , the number of Cell3 volumes along each primitive direction is $m=3^g$. Direct lengths scale as 3^g , face areas as 9^g , and volumes as 27^g . Reciprocal wavenumbers scale as 3^{-g} , while reciprocal quadratic values scale as 9^{-g} .

The merged half-step lattice contains $Q_g=(2m+1)^3$ nodes. The number of lambda sheets is $S_g=6m+1$. The central sheet contains $C_g=3m^2+3m+1$ states. The number of noncentral antipodal pairs is $(Q_g-1)/2$.

The hierarchy contains exact structural transitions. At $g=1$ the Lucas Group contains a completely surrounded central Cell3 and supports 28 independent center-graph cycles. At $g=2$ the Mass-Bridge-complete Lucas Supergroup becomes node-bulk dominated. At $g=3$ both the Cell3 volume fraction and the refined-node fraction are strongly bulk dominated, and the internal-to-exposed-interface ratio reaches 13.

Recursive closure is not approximate visual resemblance. A $9 \times 9 \times 9$ array of objects at one level reproduces the Lucas-Supergroup census at the next object scale. This exact scale recursion continues through higher orders.

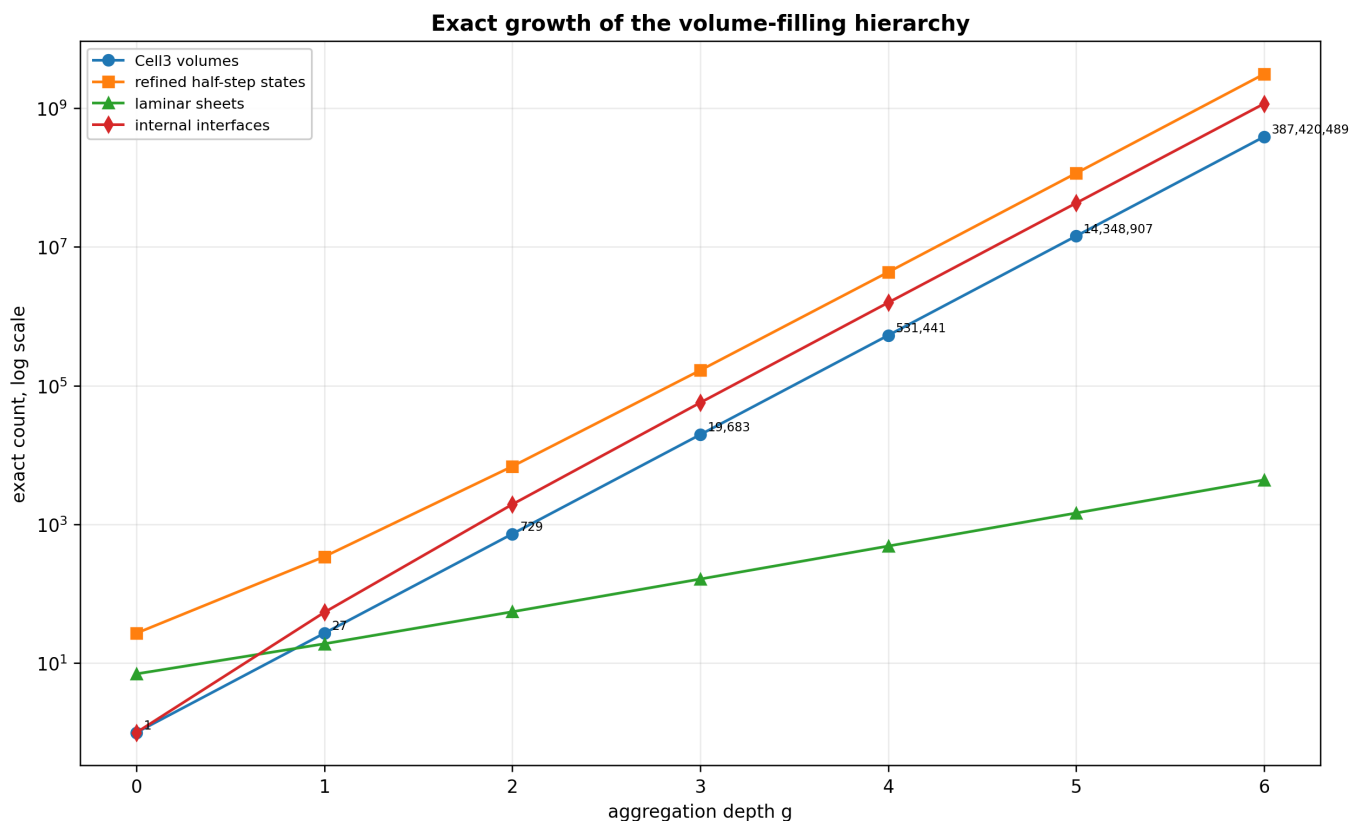


Figure 20. Exact hierarchy growth through $g=6$. Length scales as 3^g , volume count as 27^g , while the refined-state, sheet, and interface counts follow their closed formulas.

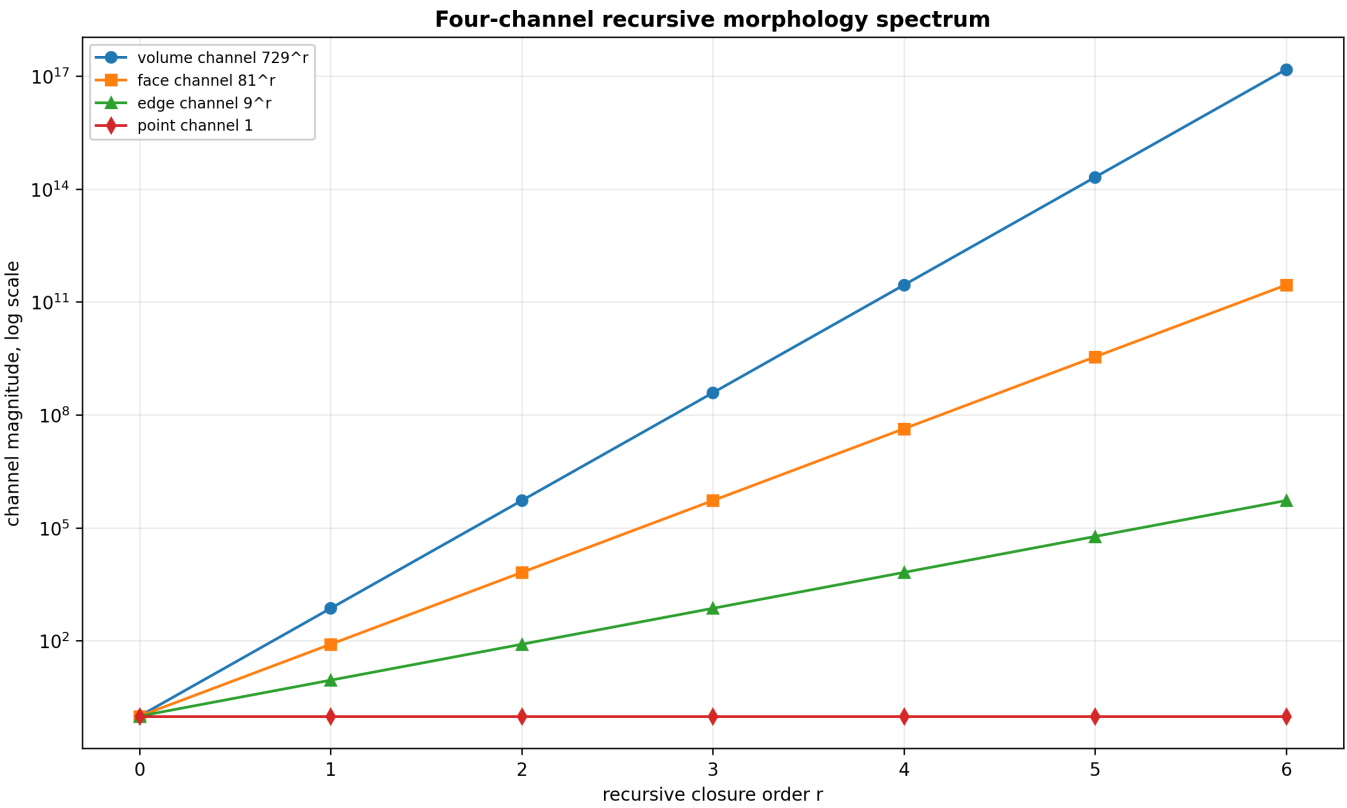


Figure 21. The exact four-channel spectrum 729^r , 81^r , 9^r , and 1 corresponding to volume, face, edge, and point contributions. Any linear observable formed from these channels obeys the associated fourth-order recurrence.

g	m	Cell3	corner nodes	half-step states	sheets	central sheet	interfaces	cycle rank
0	1	1	8	27	7	7	0	0
1	3	27	64	343	19	37	54	28
2	9	729	1,000	6,859	55	271	1,944	1,216
3	27	19,683	21,952	166,375	163	2,269	56,862	37,180
4	81	531,441	551,368	4,330,747	487	19,927	1,574,640	1,043,200
5	243	14,348,907	14,526,784	115,501,303	1,459	177,877	42,869,574	28,520,668
6	729	387,420,489	389,017,000	3,105,745,579	4,375	1,596,511	1,160,667,144	773,246,656

Four-channel recurrence

$$X_r = A 729^r + B 81^r + C 9^r + D$$

$$X_{(r+4)} = 820 X_{(r+3)} - 67158 X_{(r+2)} + 597780 X_{(r+1)} - 531441 X_r$$

19. Reciprocal Spectrum and Recursive Energy Structure

For a periodic block with outer basis mB , allowed reciprocal vectors have the form $k_q = 2\pi (mB)^{-T} q$. Their squared magnitudes are $(2\pi/m)^2 q^T R q$. Every factor of three in direct size therefore produces a factor-of-three reciprocal subharmonic.

The first Lucas Supergroup primitive reciprocal shell contains 864 antipodally identified directions, all distinct in quadratic value at twelve-decimal resolution. The minimum recorded shell value is 130.59352347783923 at direction (1,4,9); the maximum is 1744.3612920595351 at direction (9,9,-8).

The recursive energy law $E(m) = C m^2(3m-1)$ separates into volume and face channels: $E_r = 3C 729^r - C 81^r$. It therefore obeys the second-order recurrence $E_{(r+2)} = 810 E_{(r+1)} - 59049 E_r$.

The spectral hierarchy connects direct aggregation and reciprocal mode formation. Enlarging the object increases the number of collective sectors while decreasing the fundamental reciprocal spacing. Refinement performs the inverse transformation.

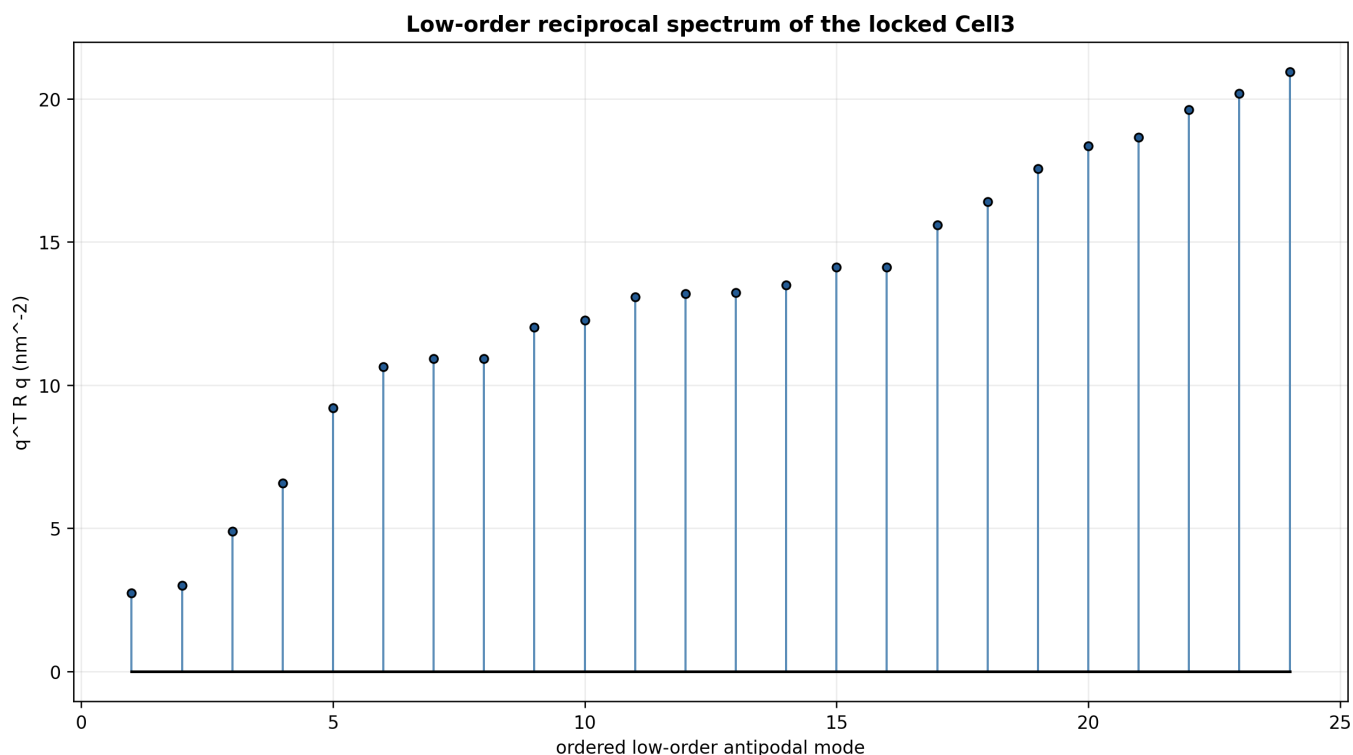


Figure 22. The first 24 low-order reciprocal quadratic values. Direct aggregation by a factor of three divides reciprocal wavenumber by three and quadratic spectral values by nine.

$$k_q = 2\pi (mB)^{-T} q = (2\pi / 3^g) B^{-T} q$$

$$|k_q|^2 = (2\pi / 3^g)^2 q^T R q$$

$$\omega(g,n)/\omega_0 = 2^{n/3^g}$$

$$E(g,n)/E_0 = 4^{n/9^g}$$

20. Physical Witness Architecture and the 29 Locks

A physical witness is a measured or exactly defined quantity used to test a model value generated earlier in the K-Field dependency chain. Witness values remain downstream: they do not determine the Cell3 basis, Mass-Bridge ratios, Devon invariants, Lucas counts, or observer state.

The first family contains 17 non-gravitational precision rows. The second contains 12 observer-rendered gravitational rows: selected G, three dimensionless gravitational couplings, and eight Planck-family quantities. The native scalar Gamma2 is not compared directly with SI G. It is first rendered through the authoritative default observer packet.

Native gravity scalar and observer rendering

$$\begin{aligned}\text{Gamma2} &= A_{ab}^9 A_{ac}^8 / (100^{11} A_{bc}^{11}) \\ &= 10^3 A_{ab}^6 \alpha_N^8 \eta_N^{-3} \\ d \ln \text{Gamma2} &= 6 d \ln A_{ab} + 8 d \ln \alpha_N - 3 d \ln \eta_N\end{aligned}$$

Object	Computation	Value	Role
Gamma2	locked native gravity scalar	2.400587821426810e-43	D08 geometric precursor
C_G_obs_SI	G_final / Gamma2 for the default packet	2.780277372245049732e32	D09 + D11 observer compiler
G_final	Gamma2 x C_G_obs_SI	6.674300000000000e-11 m^3 kg^-1 s^-2	closed D10/D11 CODATA lock

G-family computation rule

All twelve G-family witness rows are computed from G_final after the observer packet has closed the scalar G row. The three coupling rows use G_final with the CODATA electron and proton masses and exact hbar and c. The Planck rows use the standard SI formulas with G_final, exact hbar, c, and k_B. The model and witness columns therefore agree to the displayed precision.

$$\begin{aligned}\alpha_{G,e} &= G_{\text{final}} m_e^2 / (\hbar c) \\ \alpha_{G,ep} &= G_{\text{final}} m_e m_p / (\hbar c) \\ \alpha_{G,p} &= G_{\text{final}} m_p^2 / (\hbar c) \\ l_P &= \sqrt{\hbar G_{\text{final}} / c^3}; m_P = \sqrt{\hbar c / G_{\text{final}}}; F_P = c^4 / G_{\text{final}}\end{aligned}$$

The CODATA witness layer remains terminal. It verifies the observer-rendered outputs after geometry, Mass Bridge, Devon, observer state, and SI rendering have generated them; it is not allowed to back-select the upstream Cell3 geometry.

#	Row	Witness	Branch	Lock class	Observer-rendered/model value	CODATA/witness value
1	A001	Reduced mass factor μ_{ep}	non-G	selected precision witness	0.999455679424762	0.999455679424762
2	A002	Proton/electron mass ratio η	non-G	selected precision witness	1836.15267342618	1836.152673426
3	A003	Electron/proton mass ratio	non-G	selected precision witness	0.000544617021488765	0.000544617021488819
4	A006	Mass/line denominator bridge R_{acab}	non-G	selected precision witness	1.33990534199293	1.33990534199292
5	A007	Fine-structure surrogate α_N	non-G	selected precision witness	0.00729735256433073	0.00729735256433142
6	A008	Speed surrogate c_N	non-G	selected precision witness	137.035999177013	137.035999177
7	A011	Carrier/mass quotient C_m	non-G	selected precision witness	1.83615267342618	1.836152673426
8	A012	Carrier/fine quotient C_α	non-G	selected precision witness	1.37035999177013	1.37035999177
9	A014	Electron reduced Compton-like length	non-G	selected precision witness	0.00729735256433073	0.00729735256433142
10	A016	Electron full Compton-like length	non-G	selected precision witness	0.0458506184135121	0.0458506184135165
11	A018	Classical-radius-like analogue	non-G	selected precision witness	5.325135444814430e-05	5.325135444815440e-05
12	A030	Thomson-cross-section-like analogue	non-G	selected precision witness	2.375636132080060e-08	2.375636132080970e-08
13	A038	Rydberg-constant-like wavenumber	non-G	selected precision witness	0.00364867628216536	0.00364867628216571
14	A039	Hydrogen Rydberg-like wavenumber	non-G	selected precision witness	0.0036466902325926	0.00364669023259295
15	A027	Magneton ratio	non-G	selected precision witness	1836.15267342618	1836.152673426
16	A031	Vacuum-permeability-like $\mu_{0,N}$	non-G	selected precision witness	0.000106502708896289	0.000106502708896309
17	A033	Vacuum-impedance-like $Z_{0,N}$	non-G	selected precision witness	0.0145947051286615	0.0145947051286628
18	A048	Newtonian gravitational constant G	G-family	observer-rendered G lock	6.674300000000000e-11 m ³ kg ⁻¹ s ⁻²	6.674300000000000e-11 m ³ kg ⁻¹ s ⁻²
19	A050	Electron gravitational coupling $\alpha_{G,e}$	G-family	observer-rendered G -derived lock	1.751809398790771e-45	1.751809398790770e-45
20	A051	Electron-proton gravitational coupling $\alpha_{G,ep}$	G-family	observer-rendered G -derived lock	3.216589510914632e-42	3.216589510914630e-42
21	A052	Proton gravitational coupling $\alpha_{G,p}$	G-family	observer-rendered G -derived lock	5.906149429765542e-39	5.906149429765540e-39
22	A053	Planck length l_P	G-family	observer-rendered SI Planck lock	1.616255024423705e-35 m	1.616255024423710e-35 m
23	A054	Planck time t_P	G-family	observer-rendered SI Planck lock	5.391246448313604e-44 s	5.391246448313610e-44 s
24	A055	Planck mass m_P	G-family	observer-rendered SI Planck lock	2.176434342717898e-8 kg	2.176434342717900e-08 kg
25	A056	Planck/proton mass ratio	G-family	observer-rendered Planck-ratio lock	1.301211175670645e+19	1.301211175670650e+19
26	A057	Planck energy E_P	G-family	observer-rendered SI Planck lock	1.956081636698373e+9 J	1.956081636698370e+09 J
27	A058	Planck temperature T_P	G-family	observer-rendered SI Planck lock	1.416784162157343e+32 K	1.416784162157340e+32 K
28	A059	Planck force F_P	G-family	observer-rendered SI Planck lock	1.210255564338206e+44 N	1.210255564338210e+44 N
29	A060	Planck power P_P	G-family	observer-rendered SI Planck lock	3.628254904411280e+52 W	3.628254904411280e+52 W

21. Experimental Meaning and Device Classes

The K-Field becomes experimentally accessible through orientation, reversal, phase, and recursion. A static apparatus may sample one local condition, but a rotating or translating apparatus sweeps a controlled path through the K-Field geometry.

Orientation-dependent measurements compare the same apparatus at different attitudes relative to the fixed ICRS tensor. Spin-reversal measurements exchange directed traversal while preserving much of the mechanical and electromagnetic environment. Translational phase measurements track observer motion through the affine lattice. Multichannel optical measurements reconstruct reciprocal plane relationships from phase differences.

Rotating tubes, rods, rings, dual-axis masses, circularly polarized elastic oscillators, and phased arrays implement directed state-space currents. Their common design objective is to produce coherent traversal through unequal phase-cost directions while suppressing ordinary vibration and common-mode mechanical forces.

Electrical extraction geometries use the phase-cost surface differently. Constant-q contours support cost-conserving flow. The strongest scalar potential difference lies across those contours, along the surface gradient. A winding or electrode pair intended to collect potential must therefore possess a component normal to the isolines rather than lying entirely along them.

Experimental interpretation requires full vector bookkeeping. A local force, torque, voltage, or phase shift must be indexed by apparatus orientation, spin direction, observer velocity, epoch, and K-Field phase. Scalar averages can erase the directional structure that carries the signal.

Measurement state vector

```
measurement state = {  
  time, ICRS position, ICRS attitude, observer velocity,  
  apparatus phase, spin direction, and instrument state  
}
```

A K-Field experiment is incomplete when it reports only a scalar response without the orientation and phase state that produced it. Vector bookkeeping is part of the measurement definition rather than an optional correction.

22. Comparison with Established Physics

No individual mathematical ingredient of the K-Field is unfamiliar. Oblique cells and reciprocal lattices occur in crystallography. Quadratic forms occur throughout mechanics and field theory. Preferred cosmological frames occur in large-scale observations. Projective finite geometry occurs in coding and incidence theory. Entropy and path counting occur in statistical mechanics. Recursive complexes occur in topology.

The distinctive feature is their unification into one universal substrate. A material crystal has a primitive cell, but the cell is made by matter and moves with the crystal. Cell3 is prior to matter and has a fixed celestial orientation. A lattice used in numerical field theory is a computational regulator; Cell3 is physically coupled. A preferred-frame vector supplies one direction; Cell3 supplies a complete oblique triad, reciprocal plane system, discrete address phase, polarity, and recursion.

General relativity and the Standard Model ordinarily express local laws without a universal three-dimensional lattice fixed to celestial directions. The K-Field adds intrinsic spatial structure while retaining the successful local descriptions as branch expressions. The comparison is therefore not between correct and discarded physics. It is between a local effective description and a deeper substrate that organizes the states on which that description acts.

A physical affine origin is especially consequential. Preferred orientation reduces rotational symmetry. A measurable lattice phase also reduces continuous translation symmetry to a discrete subgroup. Rotation and translation then produce different experimental signatures: orientation modulation and phase migration.

Conservation laws require the substrate to participate dynamically. When matter exchanges momentum with the K-Field, the total accounting must include K-Field momentum and stress-energy. The existing geometry determines the kinematic stage; the complete action supplies the dynamical exchange law.

What is mathematically familiar and physically distinctive

Oblique lattices, reciprocal planes, quadratic forms, finite projective geometry, graph spectra, Jacobians, and recursive complexes are established mathematical structures. The distinctive physical content is their locked conjunction as one universal, ICRS-oriented, affine-phase-bearing substrate that is prior to matter and witnessed across otherwise separate physical branches.

The nearest familiar analogue is a crystal, but a material crystal is made by matter, moves with its material body, and occupies a bounded region. Cell3 is treated as the primitive local geometry of a universal field. The nearest field-theory analogue is a preferred triad or spatial condensate, but the K-Field additionally contains a Mass-Bridge-fixed oblique cell, a 27-state Lucas Limit, actual laminar sheets, a projective 13-channel decoder, the Emily spectrum, the Jack differential map, and exact Lucas recursion.

23. Present State and Research Frontier

The present K-Field framework contains a locked oblique primitive cell, exact direct and reciprocal metrics, a projective Mass Bridge, a complete 27-state local registry, a seven-sheet polarity decomposition, a 13-channel metric-plus-syndrome system, a lossless scalar address, a quadratic phase-cost tensor, an exact entropy ledger, and a recursively closed Lucas hierarchy.

The strongest structural conclusion is that these are not separate discoveries placed beside one another. They are compatible projections of one substrate. Direct geometry, reciprocal geometry, projective incidence, scalar order, entropy, polarity, and recursion each preserve different invariants of the same underlying object.

The principal theoretical frontier is the complete dynamical action. It must state how the K-Field stores energy and momentum, how matter couples to orientation and affine phase, how entropy enters metastable transition rates, how local conservation is maintained, and how established branch equations emerge in their measured limits.

The principal experimental frontier is the joint inverse problem. Time-dependent multichannel measurements should reconstruct the locked geometry, reciprocal plane families, observer attitude, observer velocity, phase migration, and path statistics in one constrained solution. A successful reconstruction would infer the substrate from its dynamic projections rather than inserting the geometry as a fixed analysis input.

The principal engineering frontier is coherent state-space circulation. Devices should maximize directed traversal of the phase-cost geometry while minimizing ordinary vibration, thermal drift, and uncontrolled environmental coupling. Reversal, attitude sweep, phase tracking, and null orientations provide the strongest internal controls.

Complete current hierarchy

The current architecture now spans the observational root, exact local geometry, algebraic reduction, minimum-complete finite graph, laminar partition, graph spectrum, perturbation response, directional phase cost, path entropy, serialized addressing, projective decoding, recursive volume filling, reciprocal modes, physical witnesses, and experimental device classes.

Galaxy-axis unit sphere -> three infinite plane systems -> native intersecting cell -> Mass-Bridge-locked Cell3 -> Devon Dimension -> Lucas Limit -> Lucas Window -> Lucas Sheets -> Emily Eigen-Spectrum, with Jack Jacobian acting across the hierarchy.

Appendix A. Locked Cell3 Numerical Ledger

Object	Value
basis a	[0.327501242055891, -0.054897007884006, -0.03748750289473] nm
basis b	[0.035330775843828, 0.207319850440192, -0.547401085388508] nm
basis c	[-0.010129317626729, -0.38978770236626 , -0.689734408367268] nm
direct metric M	[[0.11167605789605 , 0.020710333280606, 0.04393723509622], [0.020710333280606, 0.343877732392787, 0.296392758961523], [0.04393723509622 , 0.296392758961523, 0.627770610077296]] nm^2
reciprocal metric R	[[9.208025529457549e+00, 1.534802428192545e-03, -6.451880356999390e-01], [1.534802428192587e-03, 4.903398312367987e+00, -2.315175584416389e+00], [-6.451880356999389e-01, -2.315175584416389e+00, 2.731171274711695e+00]] nm^-2
volume	0.117914891911745 nm^3
face areas	{"ab": 0.19486916546048413, "ac": 0.2611062357902053, "bc": 0.3578095391285615}
opposite-face spacings	{"a": 0.32954652969544684, "b": 0.4515973797212847, "c": 0.6050977415185554}
face normals ICRS	{"bc": [-0.9959651903089612, 0.08360216250737747, -0.032619290564400036], "ac": [-0.089052594101605, -0.8665767692910323, 0.4910339483262758], "ab": [0.19409220080776704, 0.9131771688212286, 0.35837923479080463]}
laminar normal	[0.876976866375295, 0.047223124231665, -0.478206600101221]
laminar spacing	0.151272688037256 nm

Appendix B. Devon, Lucas, Emily, and Jack Formula Ledger

Layer	Core statement
Cell3	$B=[a\ b\ c];\ M=B^T B;\ R=M^{-1}$
Cofactor duality	$\text{adj}(M)_{\text{diag}}=(A_{bc}^2,A_{ac}^2,A_{ab}^2);\ R=\text{adj}(M)/V^2$
Devon	$D=(O,B,M,\text{adj}(M),R,A,V,C,S,\text{Gamma2},Q_{\text{rule}},\text{branch sockets})$
Lucas Limit	$L=\{-1,0,1\}^3;\ L =27;\ L^* =26$
Lucas Window	$p(i,j,k)=B(i,j,k)/2;\ \text{graph}=P3 \times P3 \times P3$
Lucas Sheets	$\lambda=i+j+k;\ \chi=(-1)^\lambda;\ \text{populations } 1,3,6,7,6,3,1$
Emily	$\text{spec}(A_L)=\{-3\sqrt{2},-2\sqrt{2},-\sqrt{2},0,\sqrt{2},2\sqrt{2},3\sqrt{2}\};\ \text{mult. } 1,3,6,7,6,3,1$
Jack	$J=[[-1,0,1],[1,-1,0],[17,-2,-3]];\ \det=12;\ \text{SNF}=(1,1,12)$
Phase cost	$q(n)=n^T\ G_{\text{phase } n}$
Scalar address	$\rho(k)=\sigma(k)\sqrt{k^T\ G\ k}$
Lucas hierarchy	$m=3^g;\ \text{Cell3}=m^3;\ \text{refined}=(2m+1)^3;\ \text{sheets}=6m+1$
Recursive morphology	$X_r=A729^r+B81^r+C9^r+D$

Appendix C. Glossary

Term	Definition
K-Field	The global oriented, affine, recursive physical substrate.
Cell3	The primitive oriented oblique metric-address cell.
Mass Bridge	The exact projective face-area closure fixing Cell3 shape up to scale.
Devon Dimension	The reduced algebraic object containing direct, cofactor, reciprocal, carrier, exchange, scalar, and branch structures.
Lucas Limit	The smallest finite domain preserving complete negative-center-positive structure on all three primitive axes.
Lucas Window	The explicit 27-node P3 x P3 x P3 realization at the Lucas Limit.
Lucas Sheet	One constant-lambda plane within the Lucas Window.
Devon Super Sheet	A constant-lambda plane in a refined Lucas hierarchy volume, with one checkerboard polarity.
Emily Eigen-Spectrum	The adjacency spectrum of the Lucas Window graph.
Jack Jacobian	The nondegenerate differential map from primitive-axis perturbations to Devon invariant perturbations.
Phase cost	The directional quadratic value $n^T G_{\text{phase}} n$.
Scalar serialization	A lossless one-dimensional directed address whose ordering is nonlocal relative to physical adjacency.
Lucas Group	A 3 x 3 x 3 array of 27 complete Cell3 volumes.
Lucas Supergroup	A 9 x 9 x 9 array of 729 Cell3 volumes and 27 Lucas Groups.

Foundational Conclusion

The K-Field is a physically embedded, ICRS-oriented, affine and recursive substrate. The galaxy-axis unit sphere reveals three infinite plane systems; the Mass Bridge fixes their exact Cell3 similarity geometry; Devon compresses the geometry into invariant algebra; the Lucas Limit supplies minimum-complete finite closure; Lucas Sheets organize the window and its recursive volumes; Emily resolves intrinsic graph modes; and Jack gives the differential response of the hierarchy. Phase cost, entropy, serialization, projective channels, recursive morphology, and physical witnesses are downstream expressions of this one structure.

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